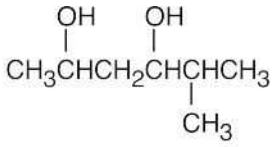
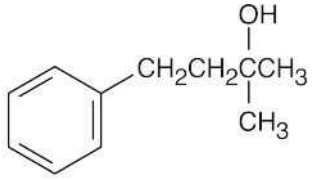
Chapter 17 – Alcohols and Phenols

Solutions to Problems

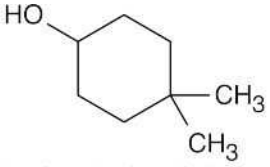
17.1 The parent chain must contain the hydroxyl group, and the hydroxyl group(s) should receive the lowest possible number.

(a) 

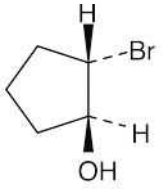
5-Methyl-2,4-hexanediol

(b) 

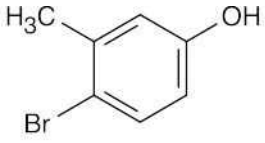
2-Methyl-4-phenyl-2-butanol

(c) 

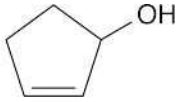
4,4-Dimethylcyclohexanol

(d) 

(1*S*,2*S*)-2-Bromocyclopentanol

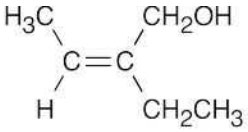
(e) 

4-Bromo-3-methylphenol

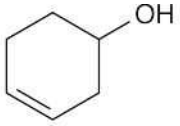
(f) 

2-Cyclopenten-1-ol

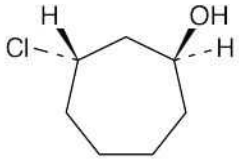
17.2

(a) 

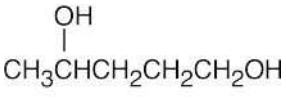
(*Z*)-2-Ethyl-2-buten-1-ol

(b) 

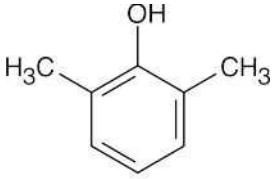
3-Cyclohexen-1-ol

(c) 

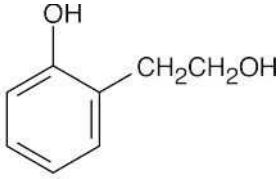
trans-3-Chlorocycloheptanol and enantiomer

(d) 

1,4-Pentanediol

(e) 

2,6-Dimethylphenol

(f) 

o-(2-Hydroxyethyl)phenol

17.3 In general, the boiling points of a series of isomers decrease with branching. The more nearly spherical a compound becomes, the less surface area it has relative to a straight chain compound of the same molecular weight and functional group type. A smaller surface area allows fewer van der Waals interactions, the weak forces that cause covalent molecules to be attracted to each other.

In addition, branching in alcohols makes it more difficult for hydroxyl groups to approach each other to form hydrogen bonds. A given volume of 2-methyl-2-propanol therefore contains fewer hydrogen bonds than the same volume of 1-butanol, and less energy is needed to break them in boiling.

17.4

(a) *Least acidic* ⟶ *Most acidic*

$HC{\equiv}CH$ < $(CH_3)_2CHOH$ < CH_3OH < $(CF_3)_2CHOH$

alkyne < hindered alcohol < alcohol < alcohol with electron-withdrawing groups

(b)

p-Methylphenol < Phenol < *p*-(Trifluoromethyl)phenol

phenol with electron-donating groups < phenol < phenol with electron-withdrawing groups

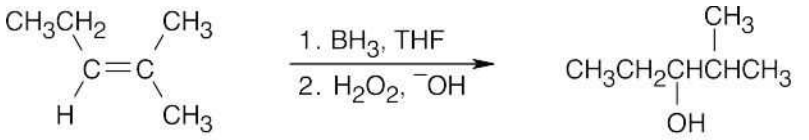
(c)

Benzyl alcohol < Phenol < *p*-Hydroxybenzoic acid

alcohol < carboxylic acid

17.5 We saw in Chapter 16 that a nitro group is electron-withdrawing. Since electron-withdrawing groups stabilize anions, *p*-nitrobenzyl alcohol is more acidic than benzyl alcohol. The methoxyl group, which is electron-donating, destabilizes an alkoxide ion, making *p*-methoxybenzyl alcohol less acidic than benzyl alcohol.

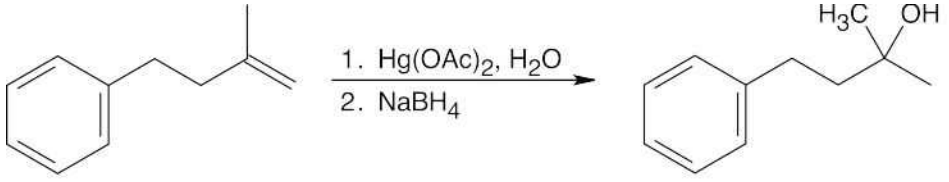
17.6 (a)



2-Methyl-3-pentanol

In a hydroboration/oxidation reaction, the hydroxyl group is bonded to the less substituted carbon.

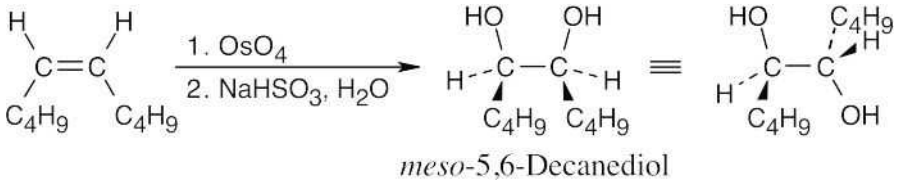
(b)



2-Methyl-4-phenyl-2-butanol

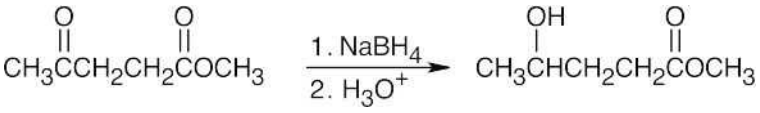
Markovnikov product results from oxymercuration/reduction

(c)



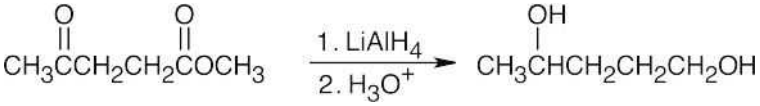
Hydroxylation results in a diol with syn stereochemistry.

17.7 (a)



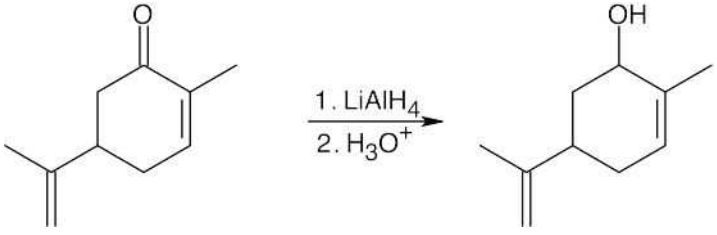
$NaBH_4$ reduces aldehydes and ketones without interfering with other functional groups.

(b)



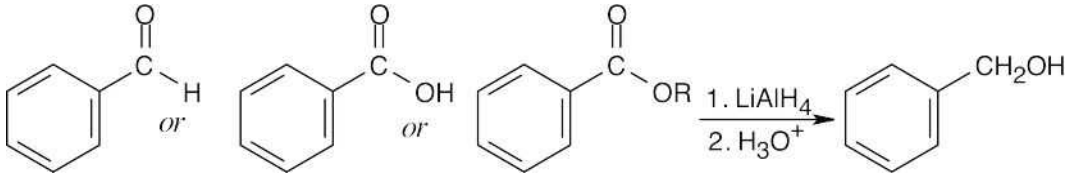
$LiAlH_4$, a stronger reducing agent, reduces both ketones and esters.

(c)



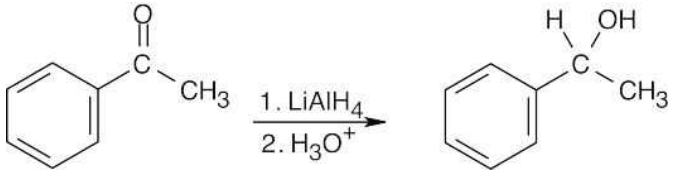
$LiAlH_4$ reduces carbonyl functional groups without reducing double bonds.

17.8 (a)



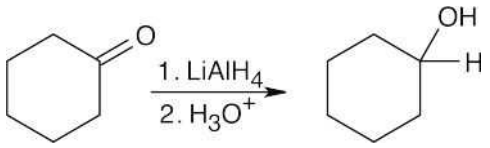
Benzyl alcohol may be the reduction product of an aldehyde, a carboxylic acid, or an ester. $NaBH_4$ may be used to reduce the aldehyde.

(b)

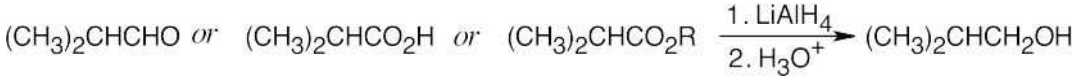


Reduction of a ketone yields the secondary alcohol. $NaBH_4$ may also be used here and in (c).

(c)

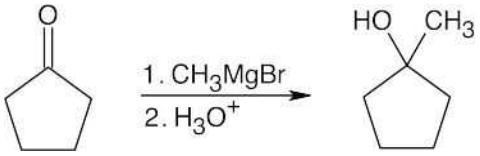


(d)

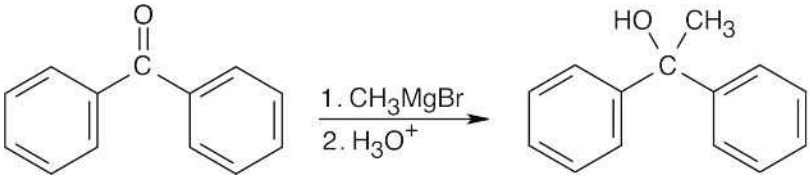


17.9 All of the products have an –OH and a methyl group bonded to what was formerly a ketone carbon.

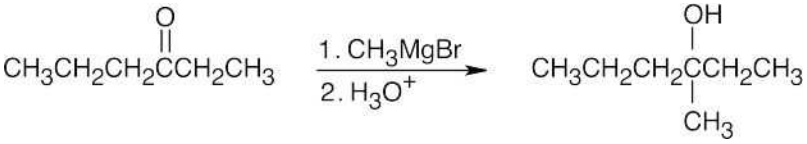
(a)



(b)

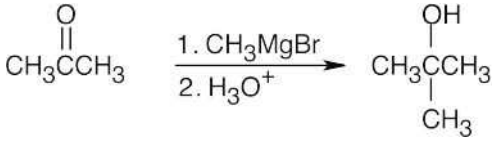


(c)

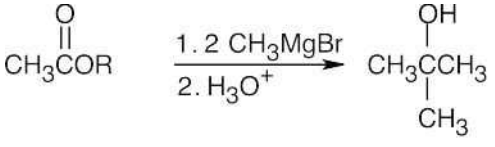


17.10 First, identify the type of alcohol. If the alcohol is primary, it can only be synthesized from formaldehyde plus the appropriate Grignard reagent. If the alcohol is secondary, it is synthesized from an aldehyde and a Grignard reagent. (Usually, there are two combinations of aldehyde and a Grignard reagent). A tertiary alcohol is synthesized from a ketone and a Grignard reagent. If all three groups on the tertiary alcohol are different, there are often three different combinations of ketone and a Grignard reagent. If two of the groups on the alcohol carbon are the same, the alcohol may also be synthesized from an ester and two equivalents of Grignard reagents.

(a) 2-Methyl-2-propanol is a tertiary alcohol. To synthesize a tertiary alcohol, start with a ketone.

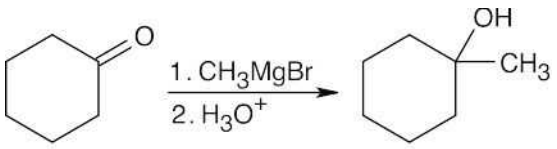


If two or more alkyl groups bonded to the carbon bearing the –OH group are the same, an alcohol can be synthesized from an ester and a Grignard reagent.



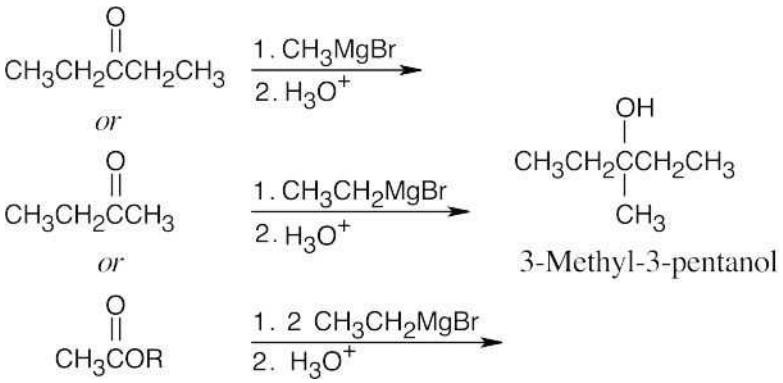
2-Methyl-2-propanol

(b) Since 1-methylcyclohexanol is a tertiary alcohol, start with a ketone.



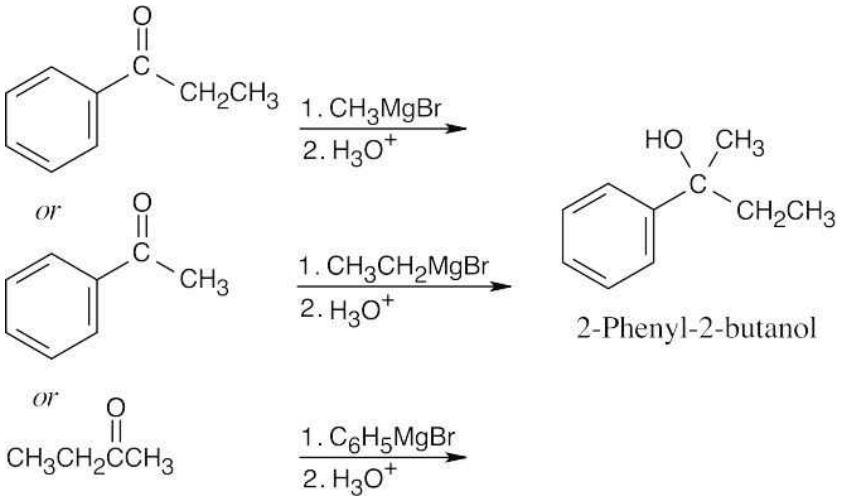
1-Methylcyclohexanol

(c) 3-Methyl-3-pentanol is a tertiary alcohol. When two of the three groups bonded to the alcohol carbon are the same, either a ketone or an ester can be used as a starting material.



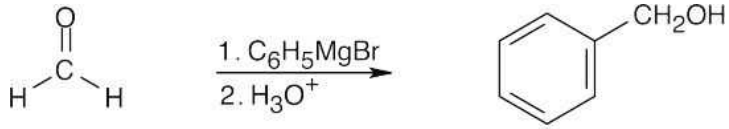
3-Methyl-3-pentanol

(d) Three possible combinations of ketone plus a Grignard reagent can be used to synthesize this tertiary alcohol.



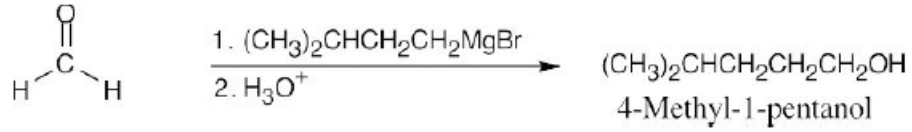
2-Phenyl-2-butanol

(e) Formaldehyde must be used to synthesize this primary alcohol.



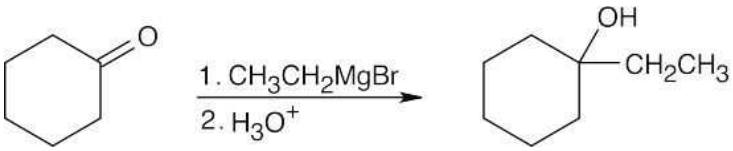
Benzyl alcohol

(f) As in (e), use formaldehyde to synthesize a primary alcohol.

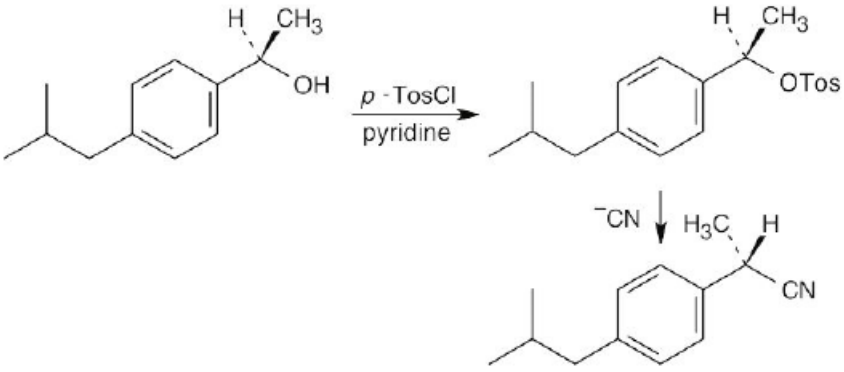


4-Methyl-1-pentanol

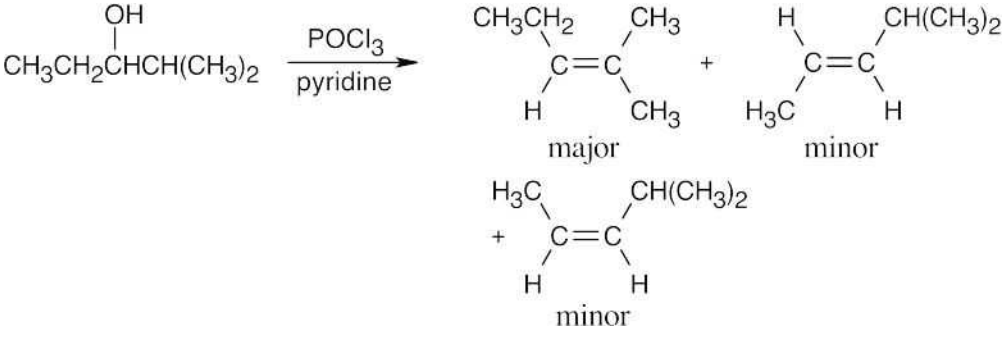
17.11 First, interpret the structure of the alcohol. This alcohol, 1-ethylcyclohexanol, is a tertiary alcohol that can be synthesized from a ketone. Only one combination of ketone and a Grignard reagent is possible.



17.12 Recall from Chapter 11 that –OH is a very poor leaving group in reactions run under S_N2 conditions. A toluenesulfonate, however, is a very good leaving group, and reaction of the toluenesulfonate of the alcohol with ^-CN proceeds readily under S_N2 conditions to give the desired product with inversion of configuration at the chirality center.

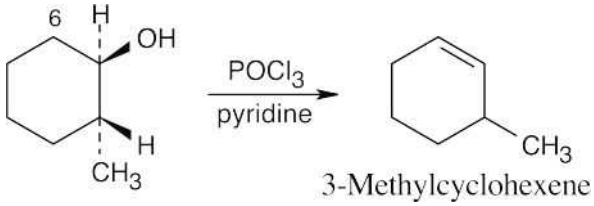


17.13 (a)



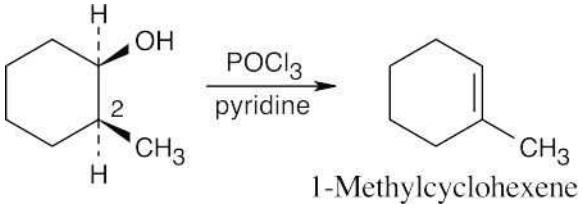
The major product has the more substituted double bond.

(b)



In E2 elimination, dehydration proceeds most readily when the two groups to be eliminated have an anti periplanar relationship. In this compound, the only hydrogen with the proper stereochemical relationship to the –OH group is at C6. Thus, the non-Zaitsev product 3-methylcyclohexene is formed.

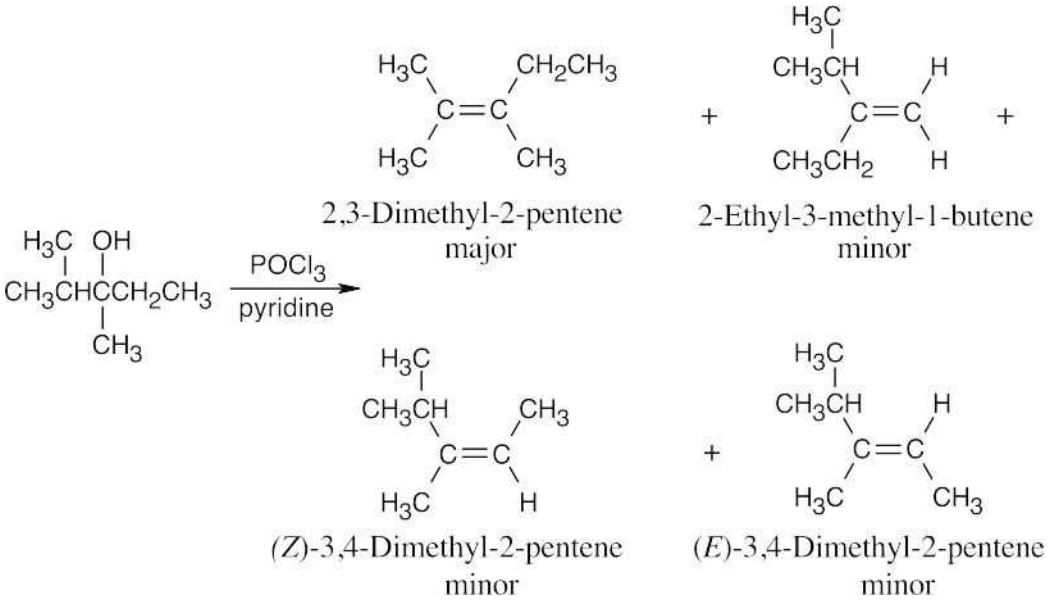
(c)



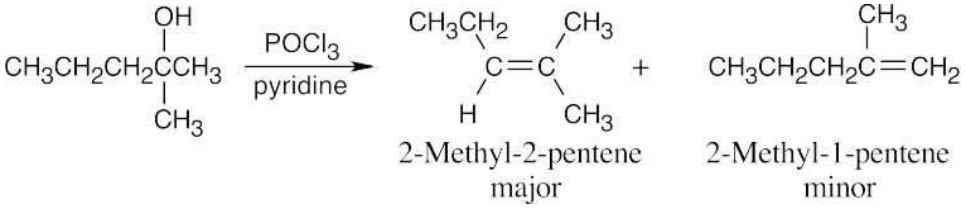
1-Methylcyclohexene

Here, the hydrogen at C2 is trans to the hydroxyl group, and dehydration yields the Zaitsev product, 1-methylcyclohexene.

(d)

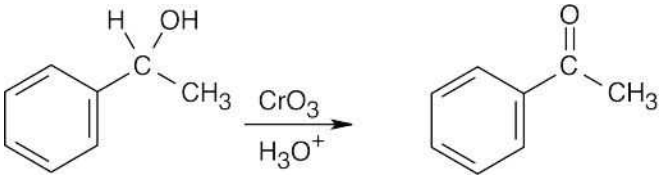


(e)

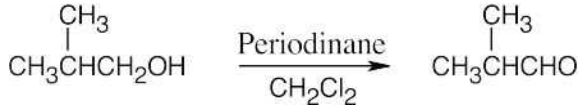


17.14 Aldehydes are synthesized from oxidation of primary alcohols, and ketones are synthesized from oxidation of secondary alcohols.

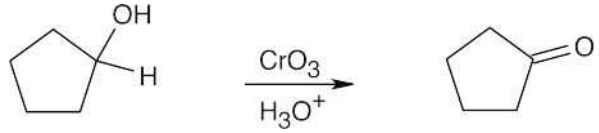
(a)



(b)



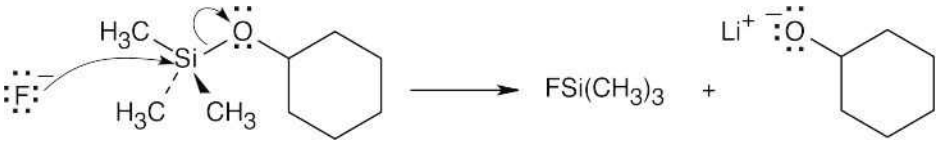
(c)



17.15

	Starting material	*CrO_3, H_3O^+ Product*	*Periodinane Product*
(a)	$CH_3CH_2CH_2CH_2CH_2CH_2OH$	$CH_3CH_2CH_2CH_2CH_2CO_2H$	$CH_3CH_2CH_2CH_2CH_2CHO$
(b)	$CH_3CH_2CH_2CH_2CH(OH)CH_3$	$CH_3CH_2CH_2CH_2C(=O)CH_3$	$CH_3CH_2CH_2CH_2C(=O)CH_3$
(c)	$CH_3CH_2CH_2CH_2CH_2CHO$	$CH_3CH_2CH_2CH_2CH_2CO_2H$	no reaction

17.16

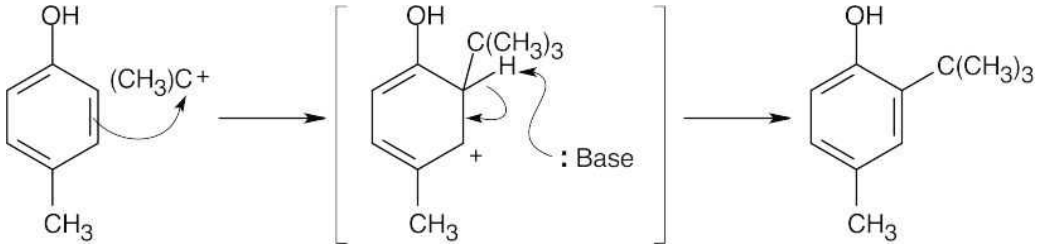


This is an S_N2 reaction in which the nucleophile F^- attacks silicon and displays an alkoxide ion as the leaving group.

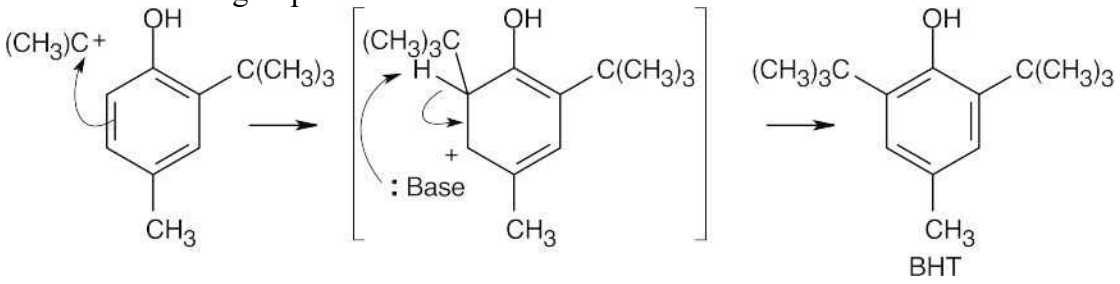
17.17



Phosphoric acid protonates 2-methylpropene, forming a *tert*-butyl carbocation.



The *tert*-butyl carbocation acts as an electrophile and alkylates *p*-cresol. Alkylation occurs ortho to the –OH group for both steric and electronic reasons.



A second *tert*-butyl carbocation alkylation forms BHT.

17.18 The infrared spectra of cholesterol and 5-cholestene-3-one each exhibit a unique absorption that makes it easy to distinguish between them. Cholesterol shows an –OH stretch at 3300-3600 cm^{-1}, and 5-cholestene-3-one shows a C=O stretch at 1715 cm^{-1}. In the oxidation of cholesterol to 5-cholestene-3-one, the –OH band disappears and is replaced by a C=O band. When oxidation is complete, no –OH absorption should be visible.

17.19 Under conditions of slow exchange, the –OH signal of a tertiary alcohol (R_3COH) is unsplit, the signal of a secondary alcohol (R_2CHOH) is split into a doublet, and the signal of a primary alcohol (RCH_2OH) is split into a triplet.

(a) 2-Methyl-2-propanol is a tertiary alcohol; its –OH signal is unsplit.

(b) Cyclohexanol is a secondary alcohol; its –OH absorption is a doublet.

(c) Ethanol is a primary alcohol; its –OH signal appears as a triplet.

(d) 2-Propanol is a secondary alcohol; its –OH absorption is split into a doublet.

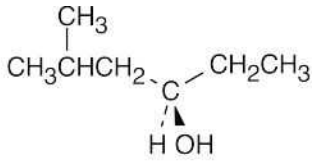
(e) Cholesterol is a secondary alcohol; its –OH absorption is split into a doublet.

(f) 1-Methylcyclohexanol is a tertiary alcohol; its –OH signal is unsplit.

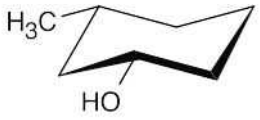
Additional Problems

Visualizing Chemistry

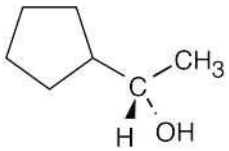
17.20

(a) 

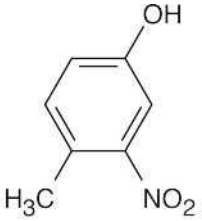
(*R*)-5-Methyl-3-hexanol

(b) 

cis-3-Methylcyclohexanol

(c) 

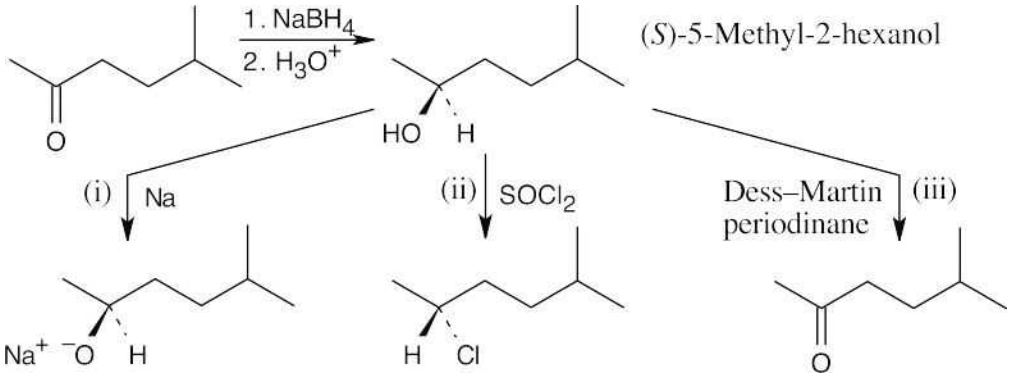
(*S*)-1-Cyclopentylethanol

(d) 

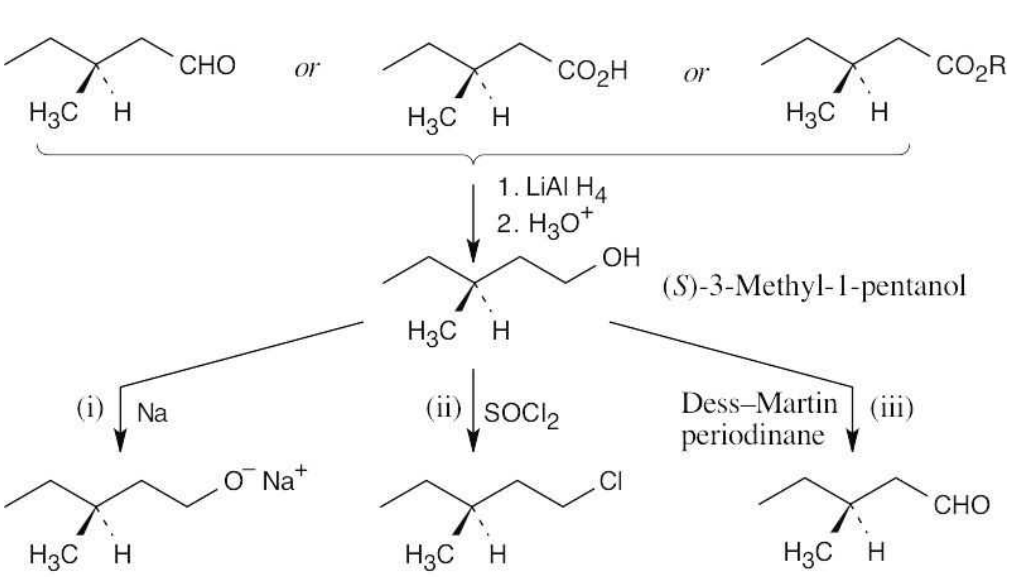
4-Methyl-3-nitrophenol

17.21 The reduction product is a racemic mixture. Reaction of the (*S*) enantiomer is shown.

(a)

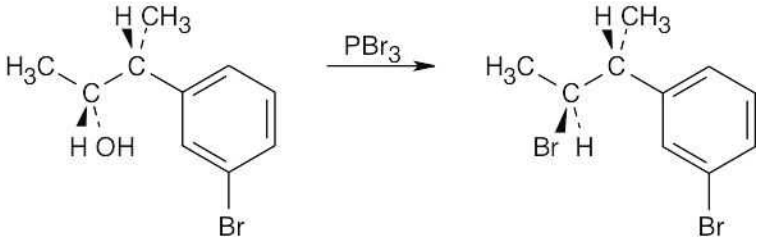


(b)

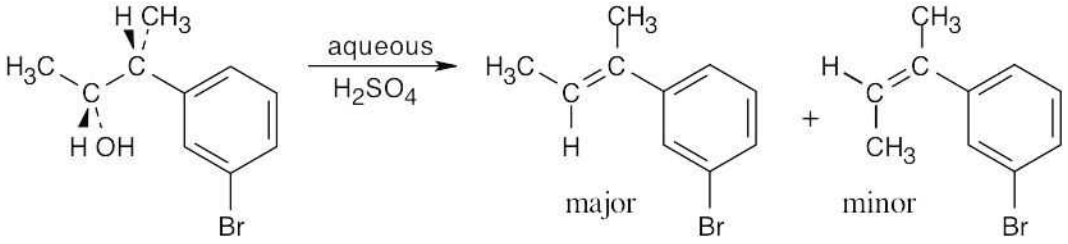

CHO
or
CO2H
or
CO2R
H3C H
1. LiAl H4
2. H3O+
OH
(S)-3-Methyl-1-pentanol
(i) Na
(ii) SOCl2
Dess–Martin periodinane
(iii)
O− Na+
Cl
CHO

17.22

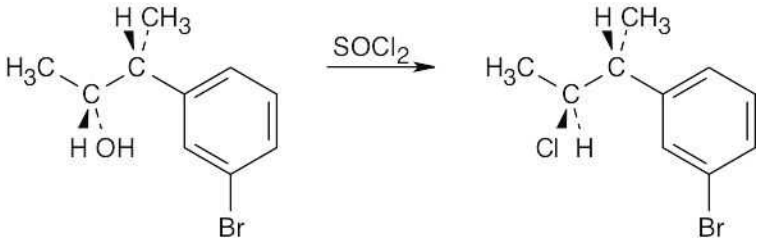
(a)


H CH3
H3C
C
H OH
Br
PBr3
Br H

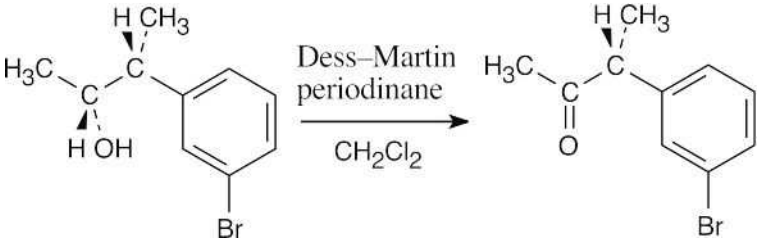
(b)


aqueous
H2SO4
CH3
H
major
minor
+

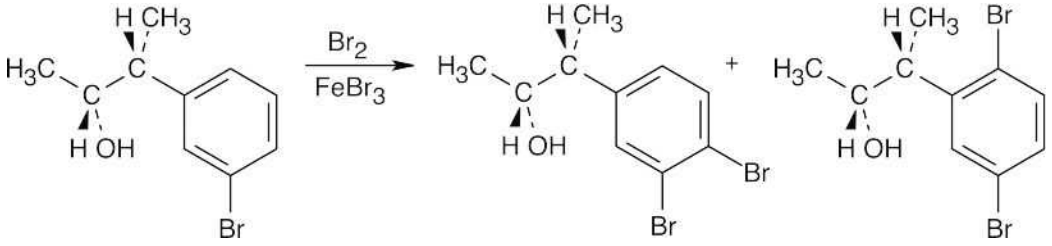
(c)


SOCl2
Cl H

(d)

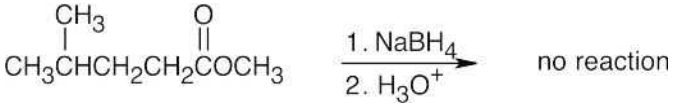

Dess–Martin periodinane
CH2Cl2
O

(e)

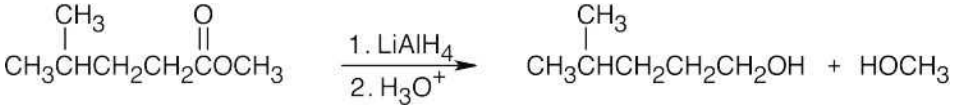


17.23

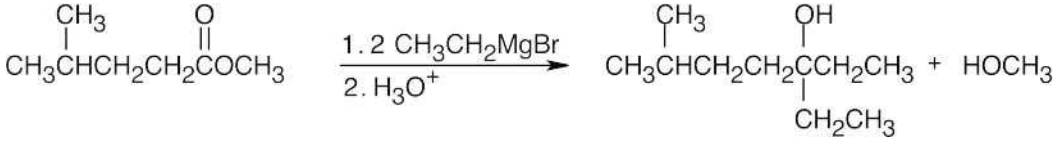
(a)



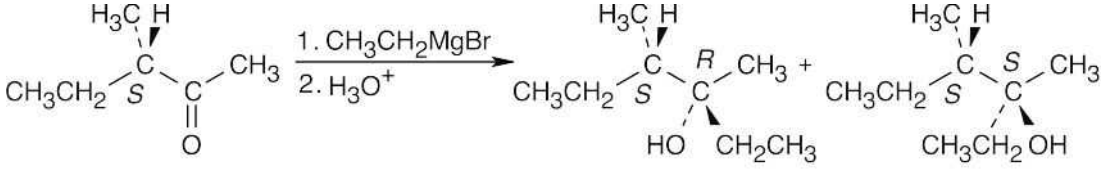
(b)



(c)



17.24



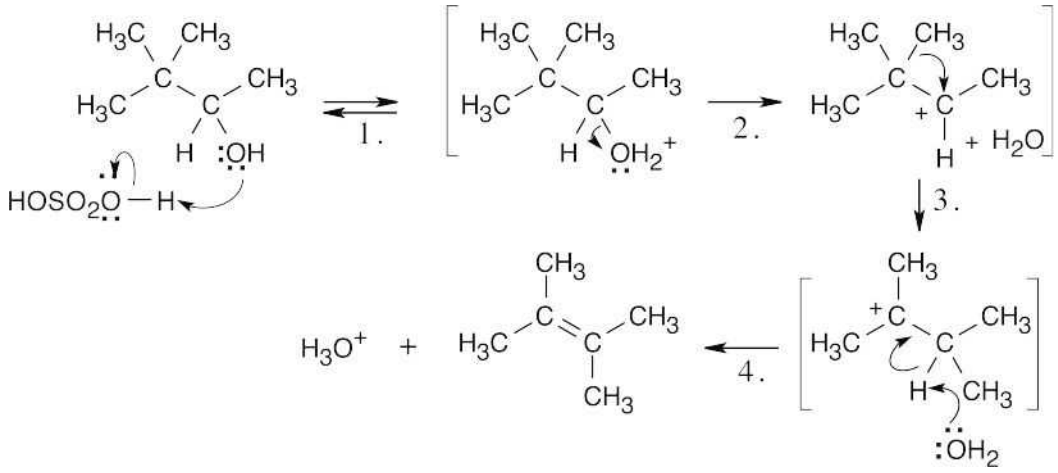
(3*R*,4*S*)-3,4-Di-methyl-3-hexanol

(3*S*,4*S*)-3,4-Di-methyl-3-hexanol

The product is a mixture of the (3*R*,4*S*) and (3*S*,4*S*) diastereomers. The diastereomers are formed in unequal amounts, and the product mixture is optically active. We can't predict which diastereomer will predominate.

Mechanism Problems

17.25



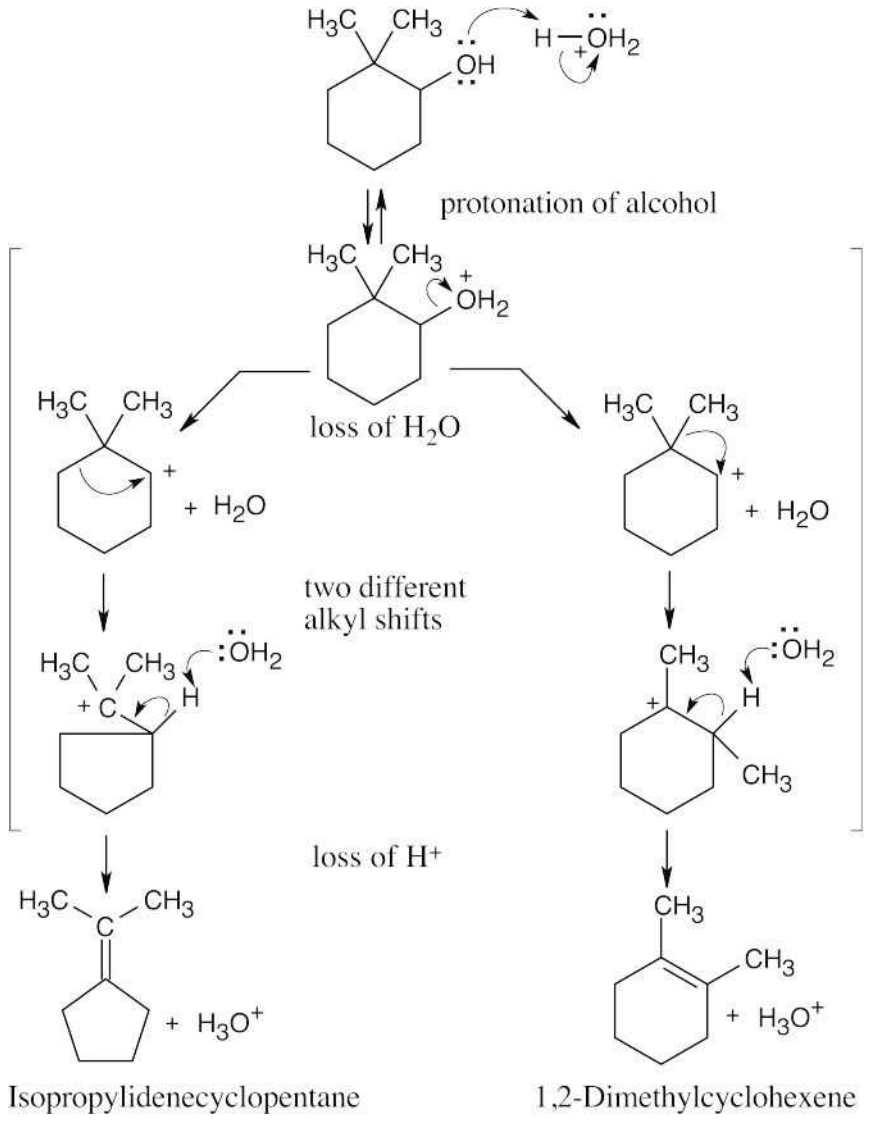
Step 1: Protonation.

Step 2: Loss of H_2O.

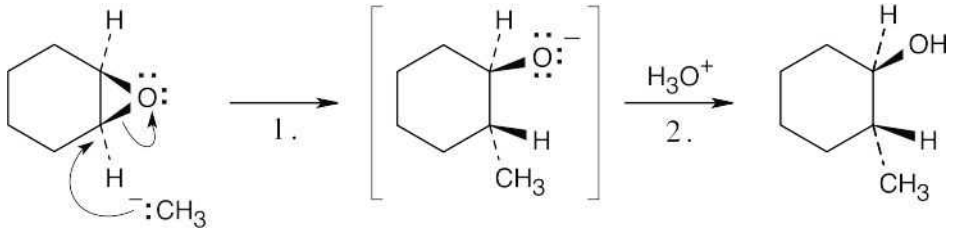
Step 3: Alkyl shift to form the tertiary carbocation.

Step 4: Loss of H_3O^+.

17.26 This mechanism consists of the same steps as are seen in Problem 17.25. Two different alkyl shifts result in two different cycloalkenes.



17.27

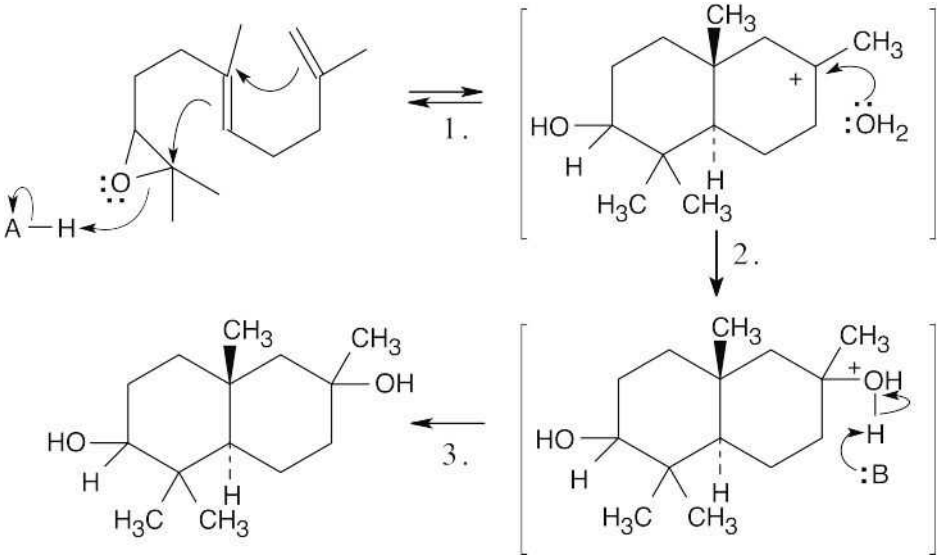


Step 1: S_N2 reaction of Grignard reagent.

Step 2: Protonation of alkoxide oxygen.

The methyl group and the hydroxyl group have a trans relationship.

17.28

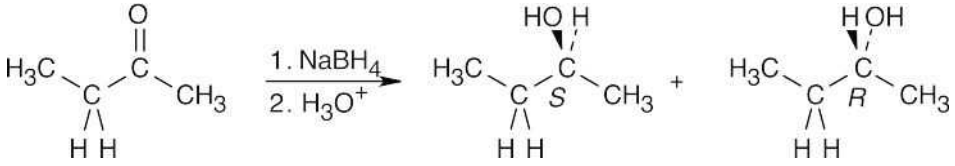


Step 1: Protonation.

Step 2: Addition of H_2O.

Step 3: Loss of H^+.

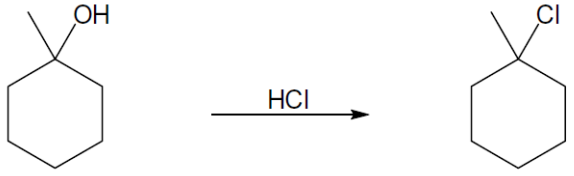
17.29



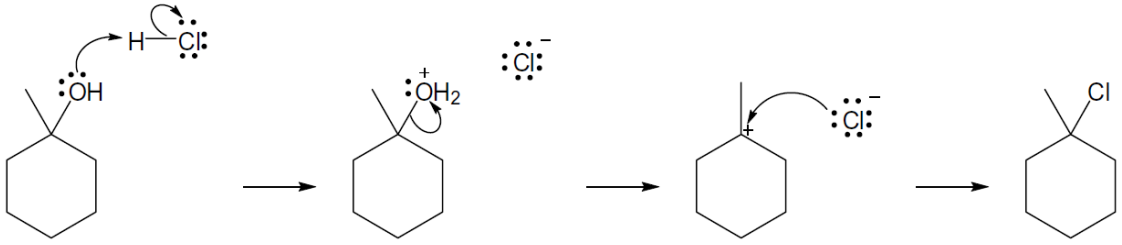
Reaction of 2-butanone with $NaBH_4$ produces a racemic mixture of (*R*)-2-butanol and (*S*)-2-butanol.

17.30

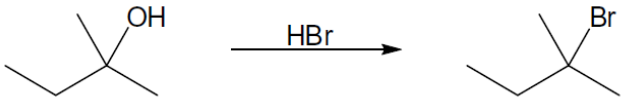
(a)



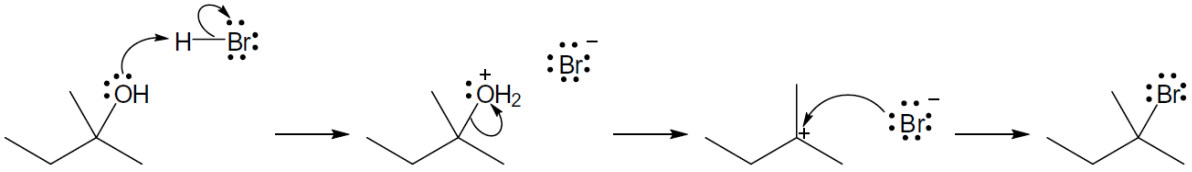
Mechanism:



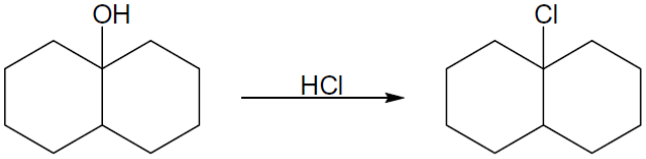
(b)



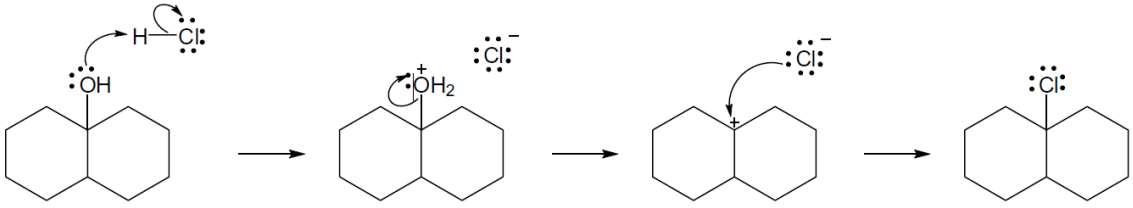
Mechanism:



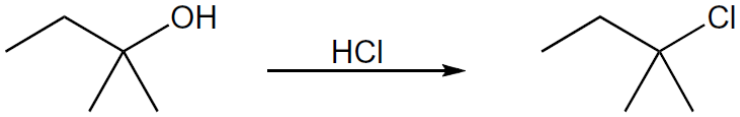
(c)



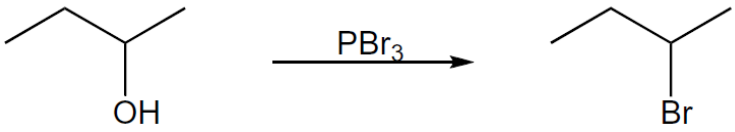
Mechanism:



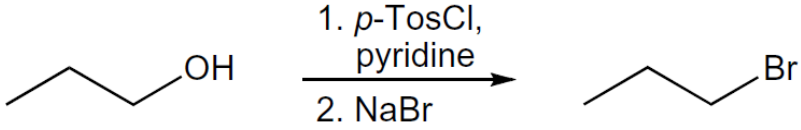
17.31 (a) Teritary alcohols undergo exclusively S_N1 substitution.



(b) Reactions with PBr_3 undergo S_N2 substitution

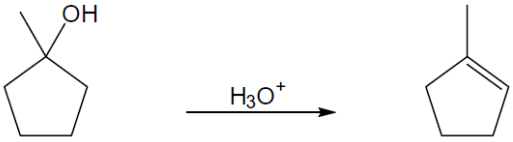


(c) In this reaction the primary alcohol is first converted to a primary tosylate. Primary tosylates undergo exclusively S_N2 substitution.

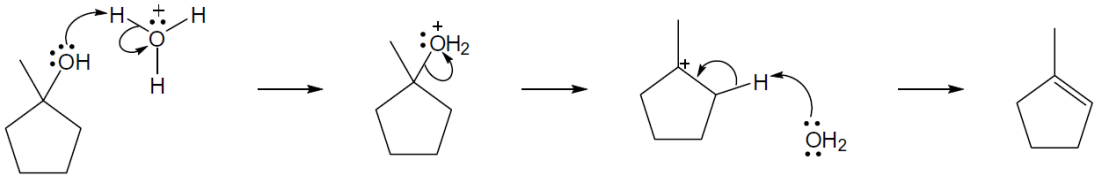


17.32

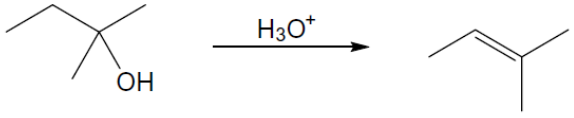
(a)



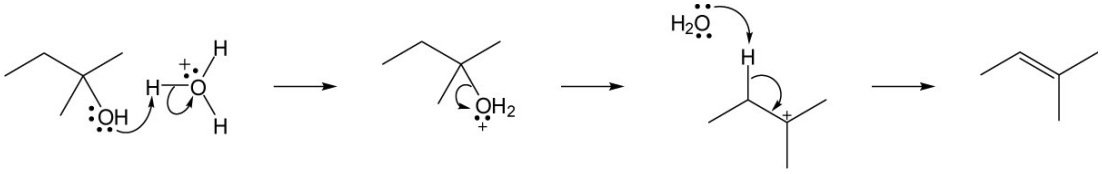
Mechanism:



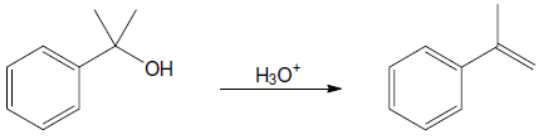
(b)



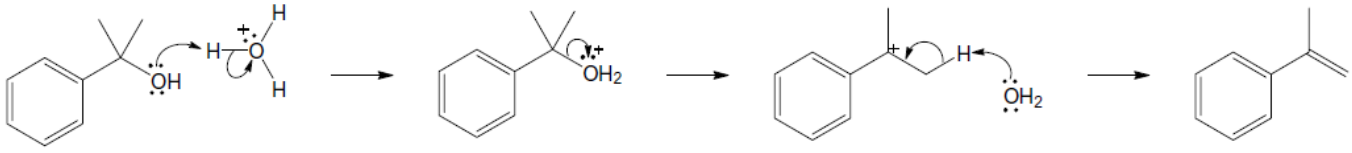
Mechanism:



(c)

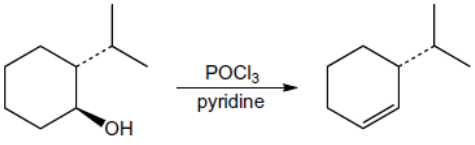


Mechanism:

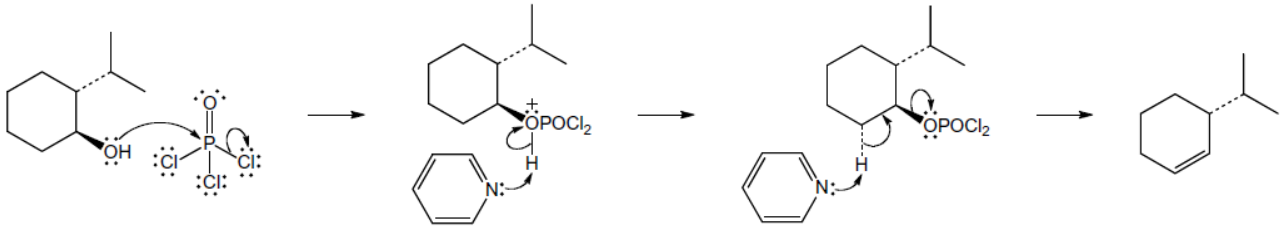


17.33

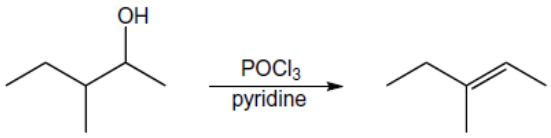
(a)



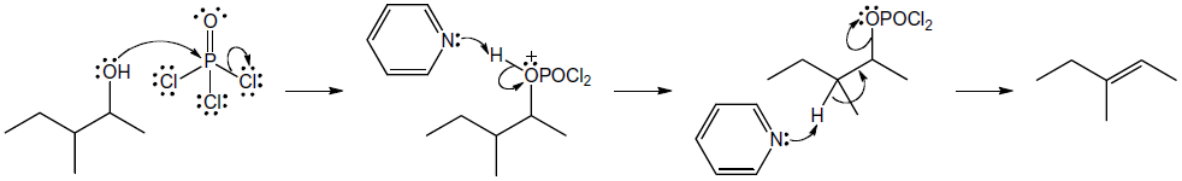
Mechanism:



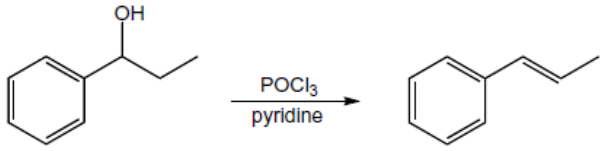
(b)



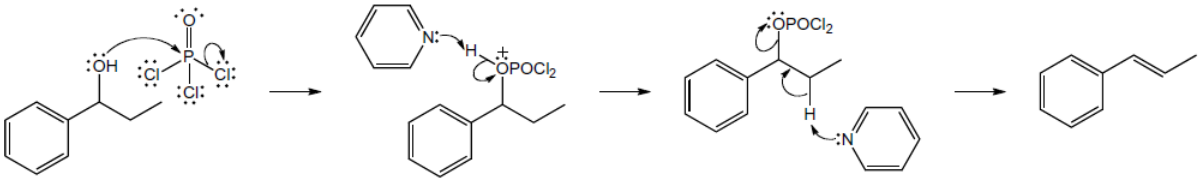
Mechanism:



(c)

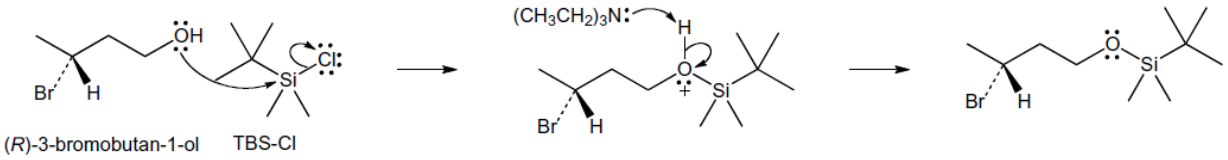


Mechanism:

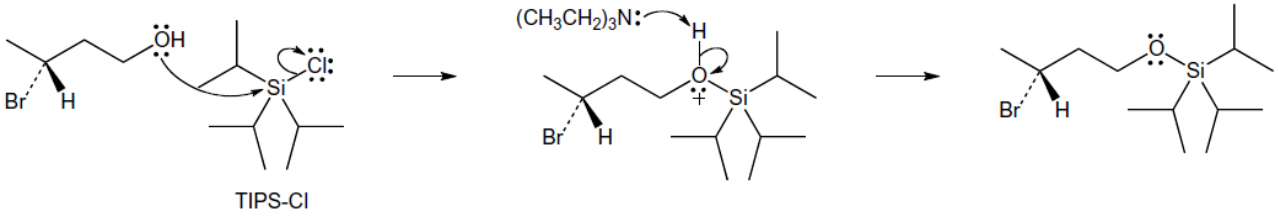


17.34

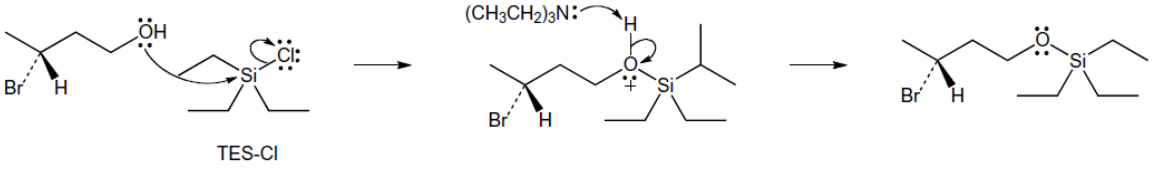
(a) *tert*-butyldimethylsilyl chloride (TBS-Cl)



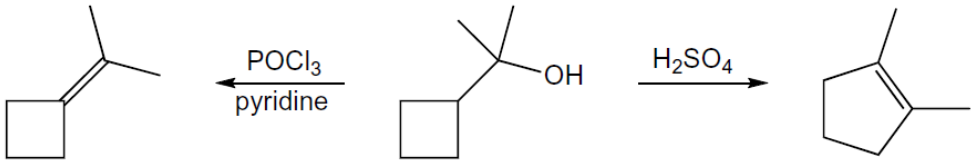
(b) triisopropylsilyl chloride (TIPS-Cl)



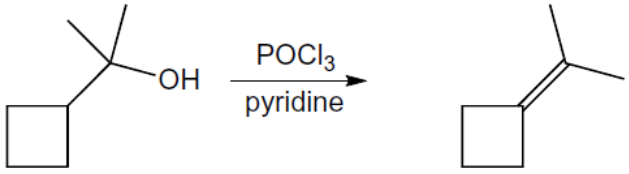
(c) triethylsilyl chlorides (TES-Cl)



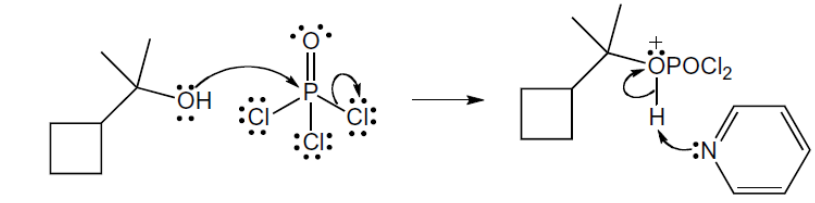
17.35

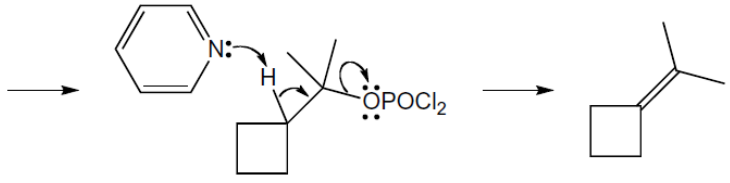


In the $POCl_3$ reaction, the elimination reaction mechanism is E2.

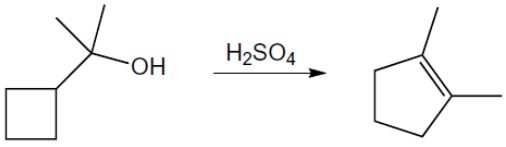


Mechanism:

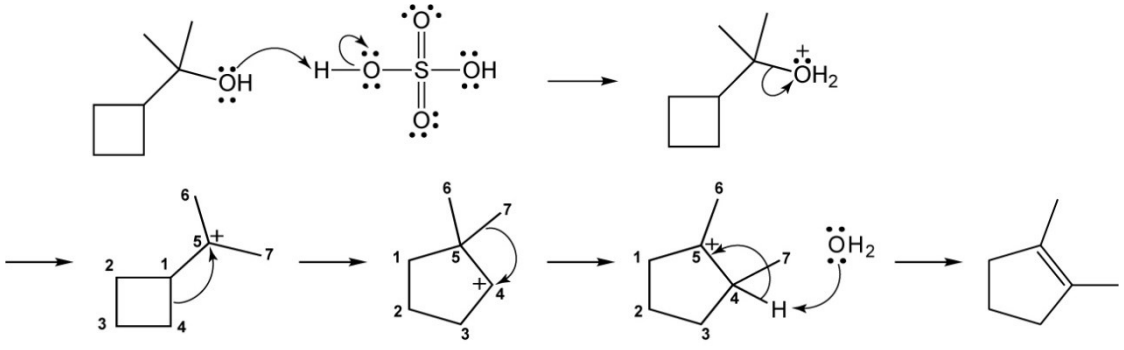




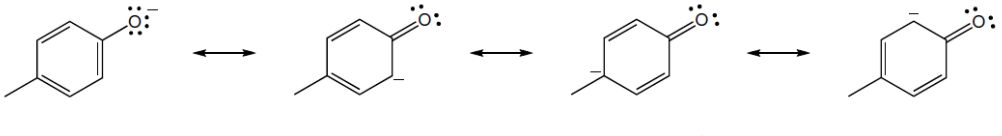
In the elimination catalyzed by sulfuric acid, the mechanism is E1 and results in a carbocation that undergoes rearrangement.



Mechanism:

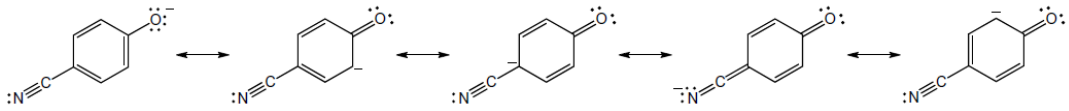


17.36 (a)



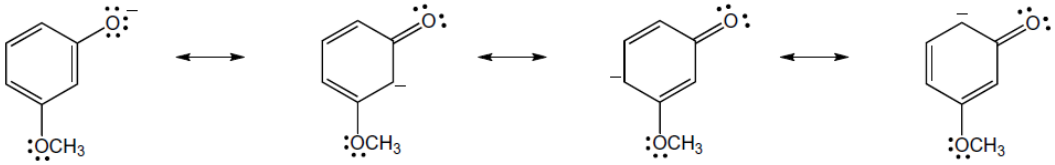
This resonance form destabilized by having negative charge next to an electron donating methyl group

(b)



This resonance form is stabilized by having the negative charge on the electronegative nitrogen

(c)

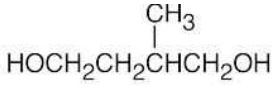


Because there is no direct effect through resonance, the OCH_3 group does not have a significant effect on the pK_a of the parent phenol

Naming Alcohols

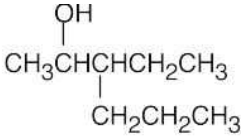
17.37

(a)



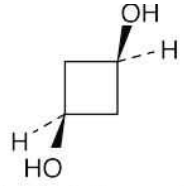
2-Methyl-1,4-butanediol

(b)



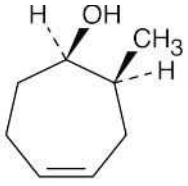
3-Ethyl-2-hexanol

(c)



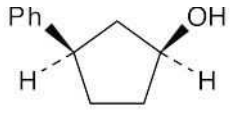
cis-1,3-Cyclobutanediol

(d)



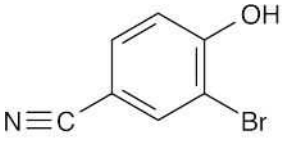
cis-2-Methyl-4-cyclohepten-1-ol

(e)



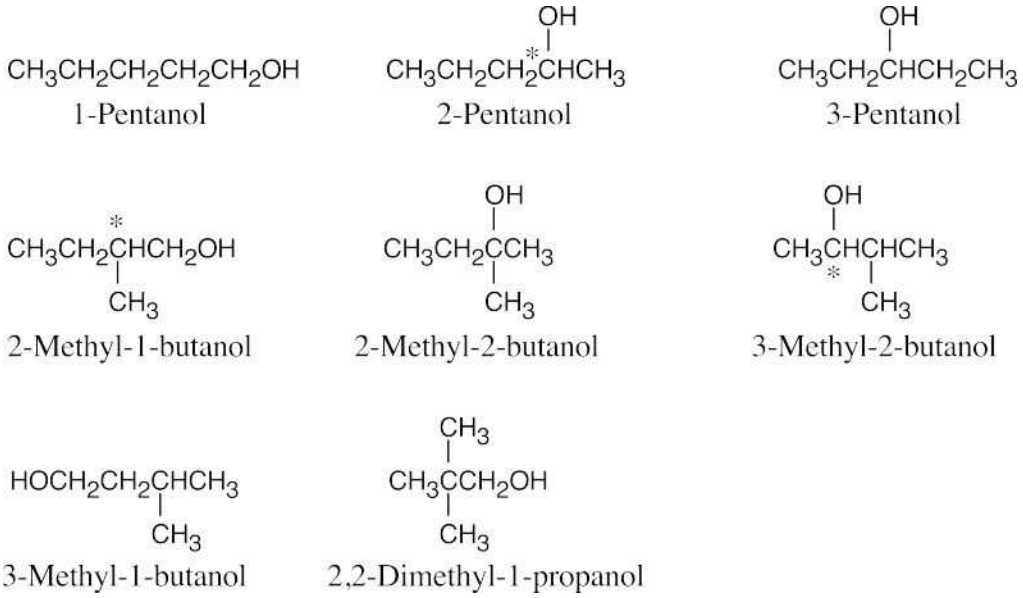
cis-3-Phenylcyclopentanol

(f)



2-Bromo-4-cyanophenol
or
3-Bromo-4-hydroxy-benzonitrile

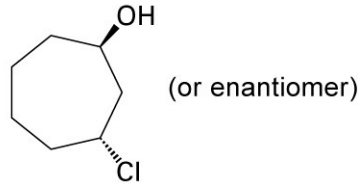
17.38 None of these alcohols has multiple bonds or rings.



2-Pentanol, 2-methyl-1-butanol and 3-methyl-2-butanol have chiral carbons (starred).

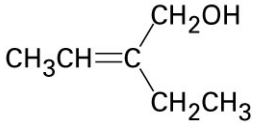
17.39

(a) *Trans*-3-Chlorocycloheptanol

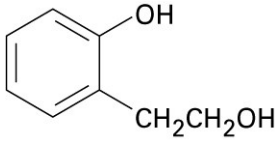


(or enantiomer)

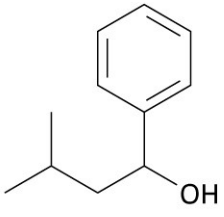
(b) 2-Ethyl-2-buten-1-ol



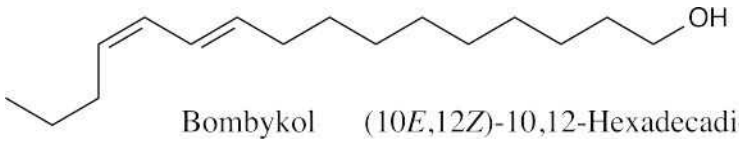
(c) *o*-(2Hydroxyethyl)phenol



(d) 3-Methyl-1-phenyl-1-butanol

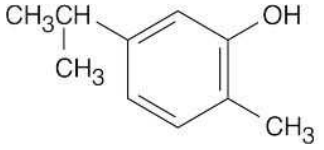


17.40



Bombykol (10*E*,12*Z*)-10,12-Hexadecadien-1-ol

17.41

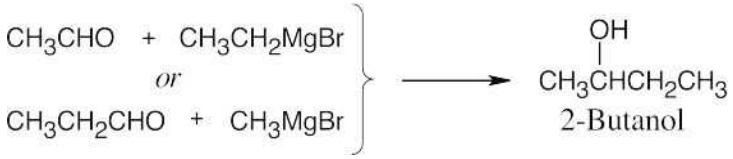


Carvacrol 5-Isopropyl-2-methylphenol

Synthesizing Alcohols

17.42 In some of these problems, different combinations of a Grignard reagent and a carbonyl compound are possible. Remember that aqueous acid is added to the initial Grignard adduct to yield the alcohol.

(a)



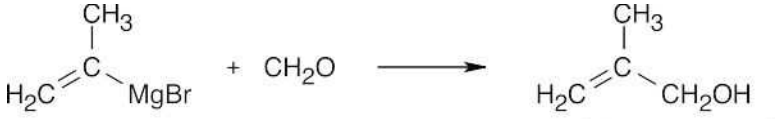
2-Butanol

(b)



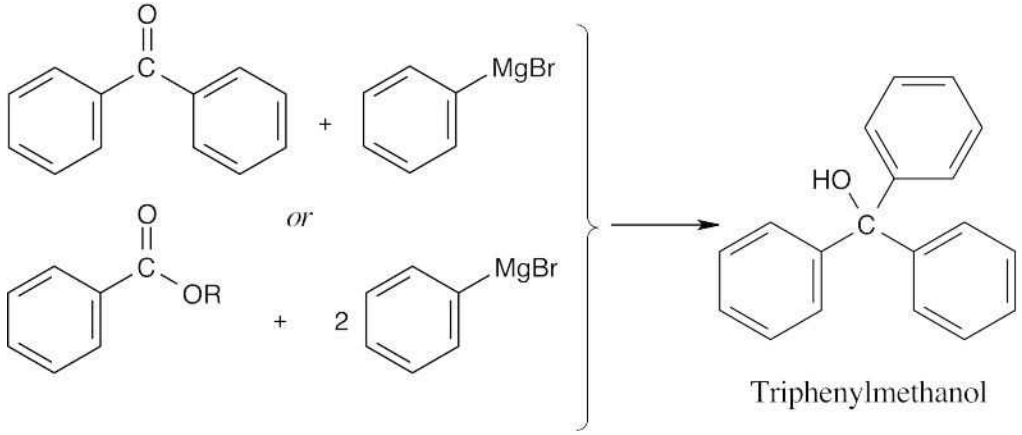
3-Pentanol

(c)



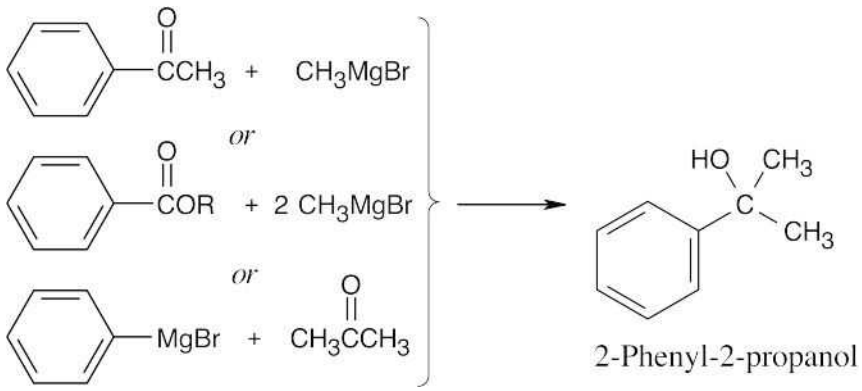
2-Methyl-2-propen-1-ol

(d)

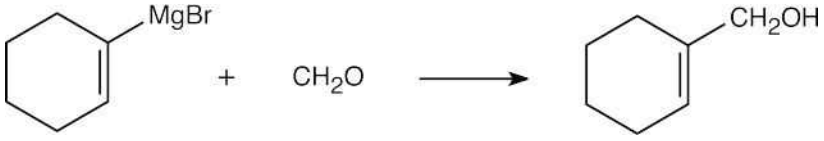


Triphenylmethanol

(e)



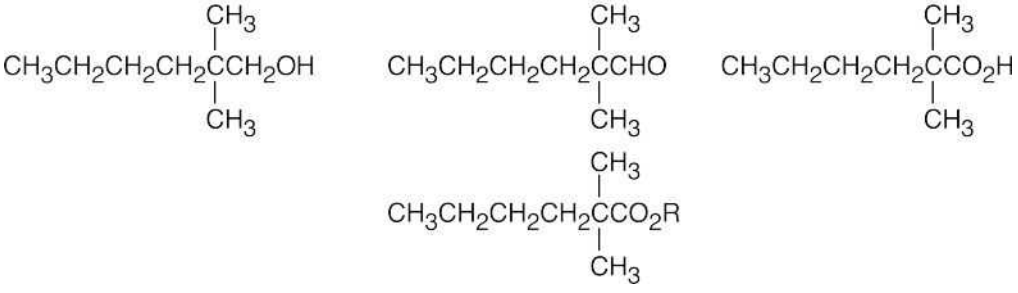
(f)



17.43

Alcohol	*Carbonyl precursor(s)*

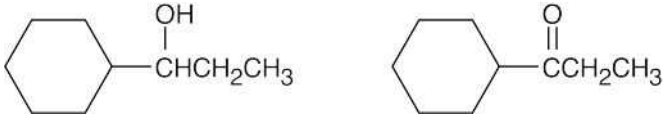
(a)



(b)



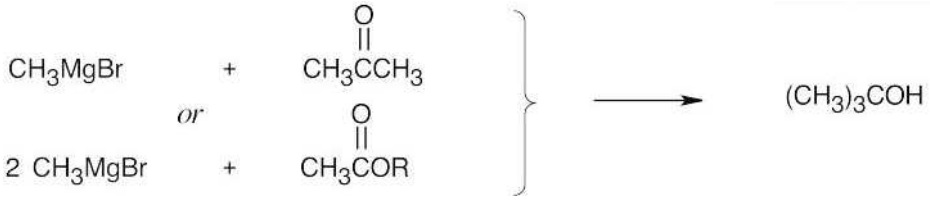
(c)



17.44

Grignard Reagent + Carbonyl Compound ⟶ *Product (after dilute acid workup)*

(a)

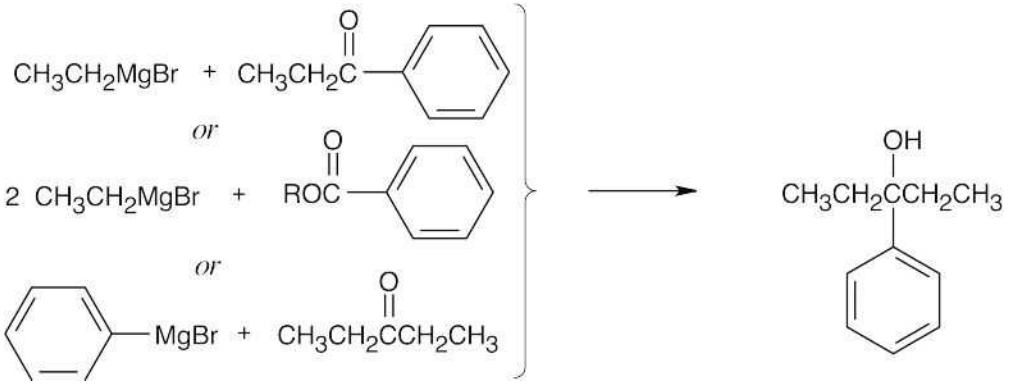


Grignard Reagent + Carbonyl Compound ⟶ *Product (after dilute acid workup)*

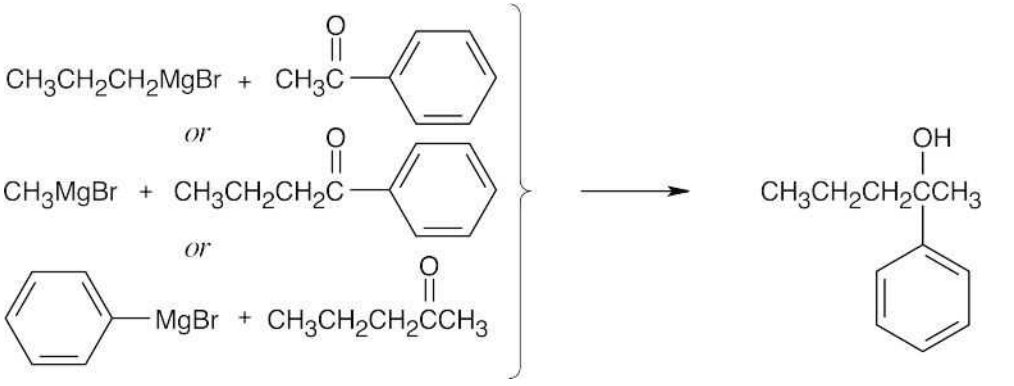
(b)



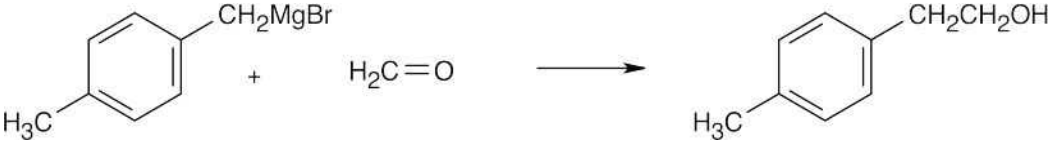
(c)



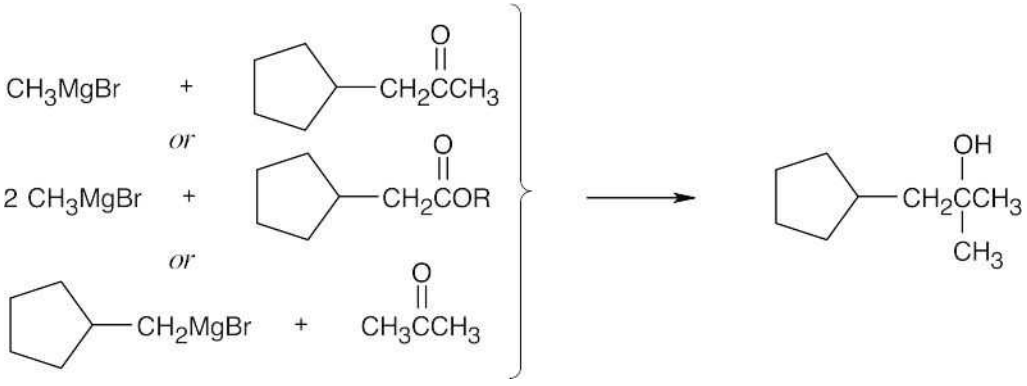
(d)



(e)

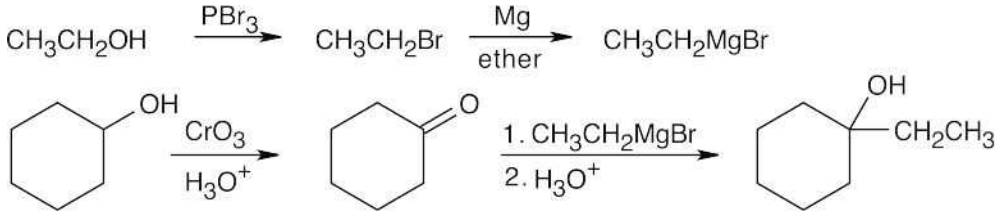


(f)

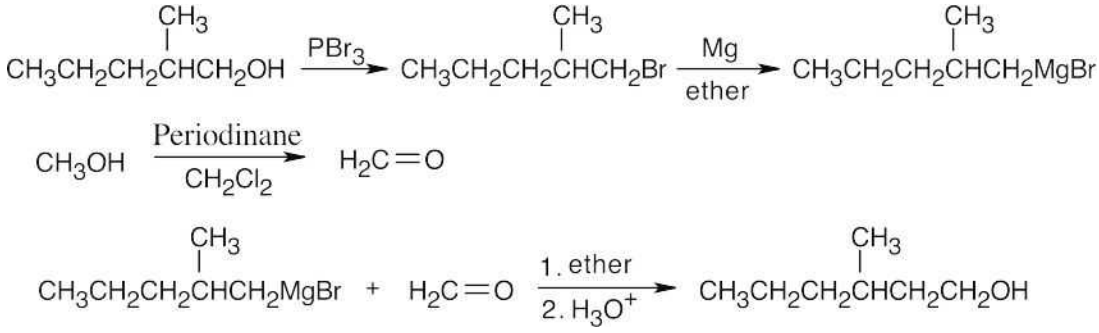


17.45 All of these syntheses involve a Grignard reaction at some step. Both the carbonyl compound and the Grignard reagent must be prepared from alcohols.

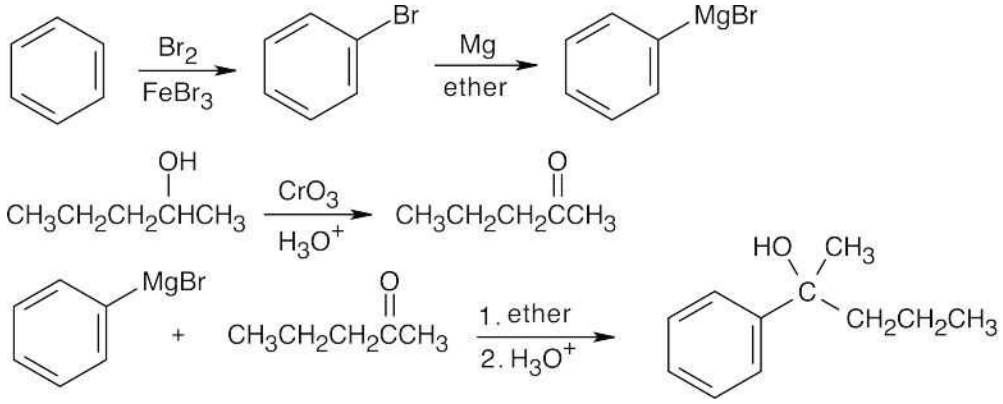
(a)



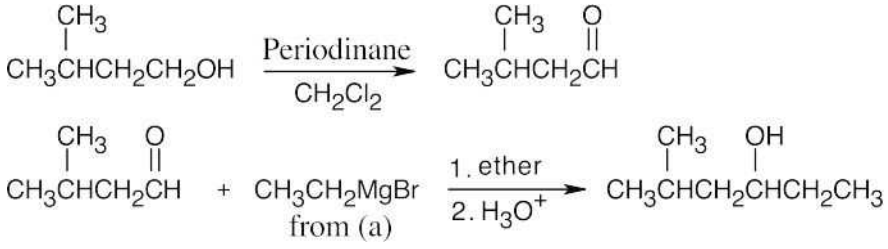
(b)



(c)



(d)



Reactions of Alcohols

17.46

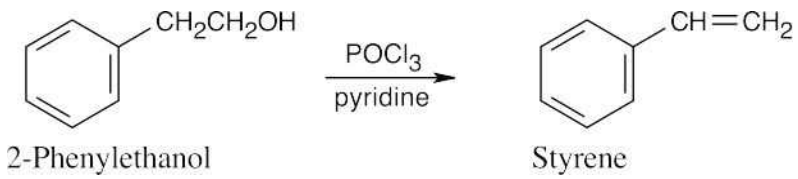
(a) $CH_3CH_2CH_2CH_2CH_2OH \xrightarrow{PBr_3} CH_3CH_2CH_2CH_2CH_2Br$

(b) $CH_3CH_2CH_2CH_2CH_2OH \xrightarrow{SOCl_2} CH_3CH_2CH_2CH_2CH_2Cl$

(c) $CH_3CH_2CH_2CH_2CH_2OH \xrightarrow[CH_2Cl_2]{\text{Periodinane}} CH_3CH_2CH_2CH_2CHO$

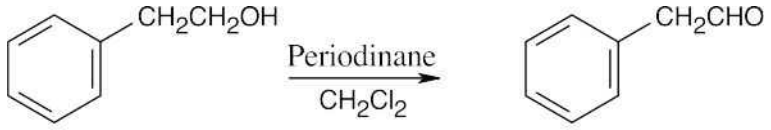
17.47

(a)



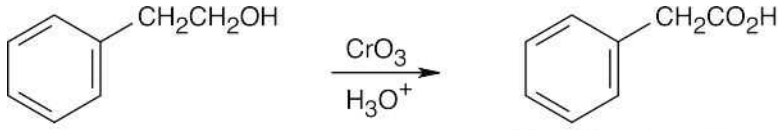
2-Phenylethanol — Styrene

(b)



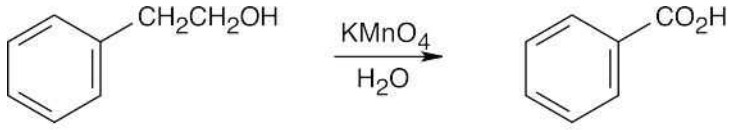
Phenylacetaldehyde

(c)



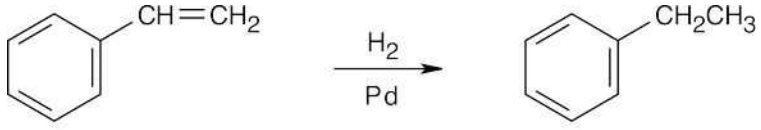
Phenylacetic acid

(d)



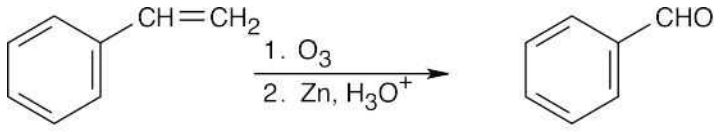
Benzoic acid

(e)



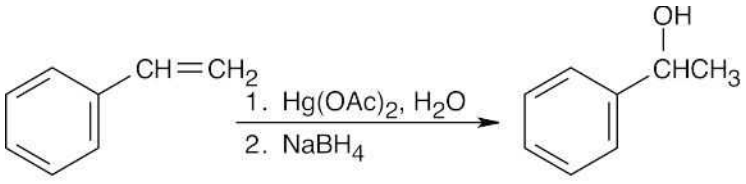
from (a) — Ethylbenzene

(f)



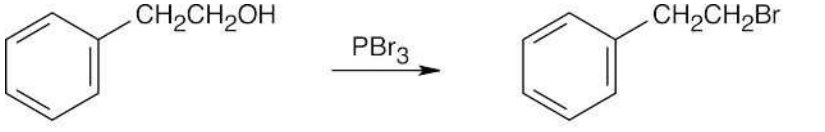
from (a) — Benzaldehyde

(g)



from (a) — 1-Phenylethanol

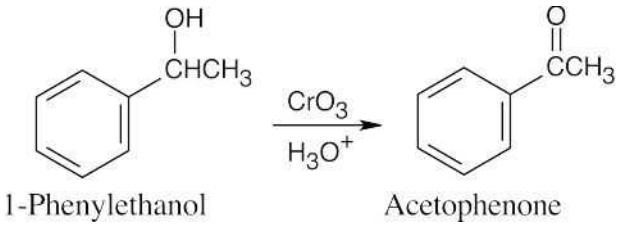
(h)



1-Bromo-2-phenylethane

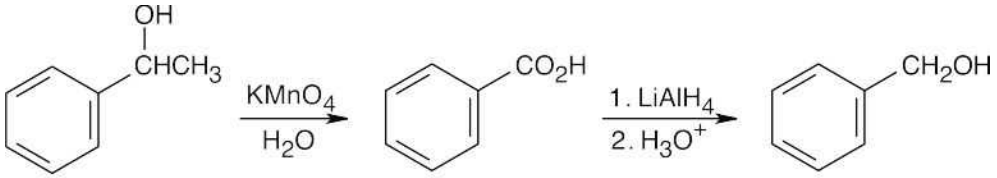
17.48

(a)



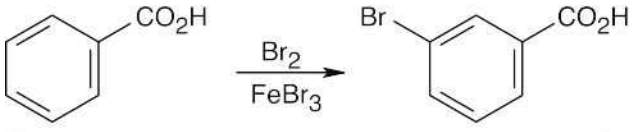
1-Phenylethanol Acetophenone

(b)



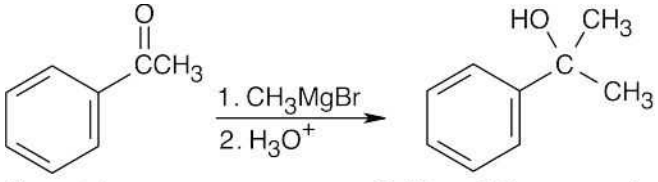
Benzyl alcohol

(c)



from (b) *m*-Bromobenzoic acid

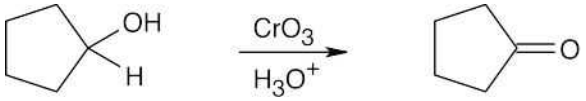
(d)



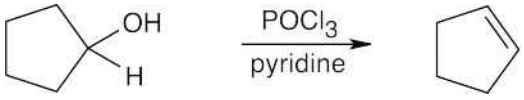
from (a) 2-Phenyl-2-propanol

17.49

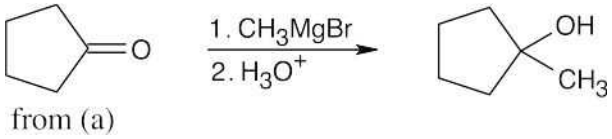
(a)



(b)

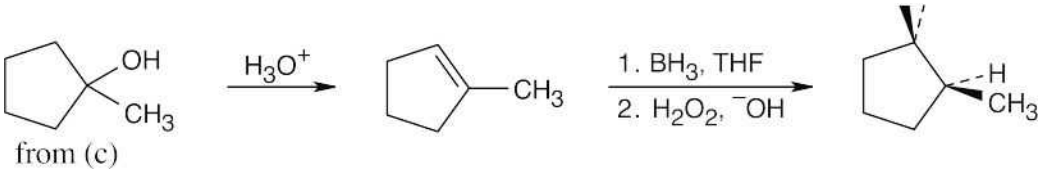


(c)



from (a)

(d)

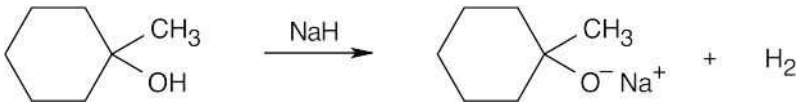


from (c)

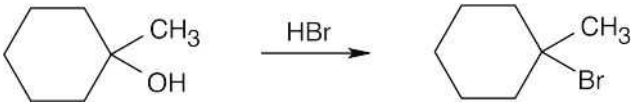
Remember that hydroboration proceeds with syn stereochemistry, and the –H and –OH added have a cis relationship.

17.50

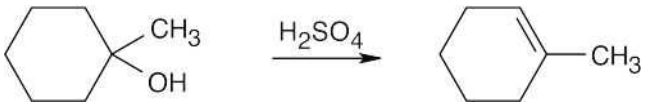
(a)



(b)

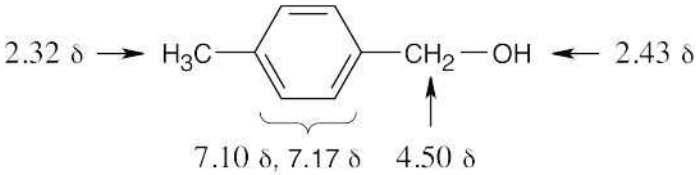


(c)



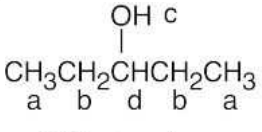
Spectroscopy

17.51



p-Methylbenzyl alcohol

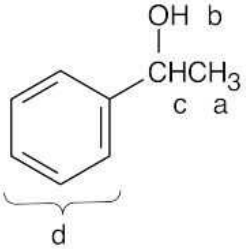
17.52 (a)



3-Pentanol

a = 0.93 δ
b = 1.42 δ
c = 1.83 δ
d = 3.41 δ

(b)

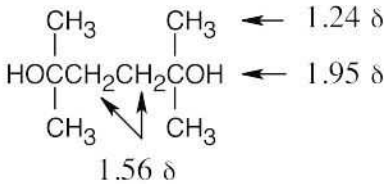


1-Phenylethanol

a = 1.42 δ
b = 2.43 δ
c = 4.80 δ
d = 7.32 δ

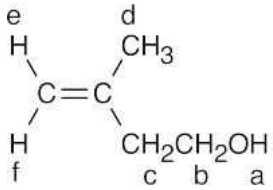
17.53 1. $C_8H_{18}O_2$ has *no* double bonds or rings, based on degree of unsaturation.

2. The IR band at 3350 cm^{-1} shows the presence of a hydroxyl group.
3. The compound is symmetrical (simple NMR).
4. There is no splitting.



2,5-Dimethyl-2,5-hexanediol

17.54



3-Methyl-3-buten-3-ol

The peak absorbing at 1.76 δ (3 H) is due to the d protons. This peak, which occurs in the allylic region of the spectrum, is unsplit.

The peak absorbing at 2.13 δ (1 H) is due to the –OH proton a.

The peak absorbing at 2.30 δ (2 H) is due to protons c. The peak is a triplet because of splitting by the adjacent b protons.

The peak absorbing at 3.72 δ (2 H) is due to the b protons. The adjacent oxygen causes the peak to be downfield, and the adjacent –CH_2– group splits the peak into a triplet.

The peaks at 4.79 δ and 4.85 δ (2 H) are due to protons e and f.

17.55 (a) $C_5H_{12}O$, $C_4H_8O_2$, $C_3H_4O_3$

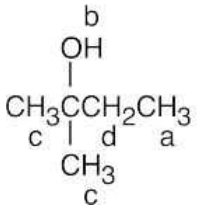
(b) The ^{1}H NMR data show that the compound has twelve protons.

(c) The IR absorption at 3600 cm^{-1} shows that the compound is an alcohol.

(d) The compound contains five carbons, two of which are identical.

(e) $C_5H_{12}O$ is the molecular formula of the compound.

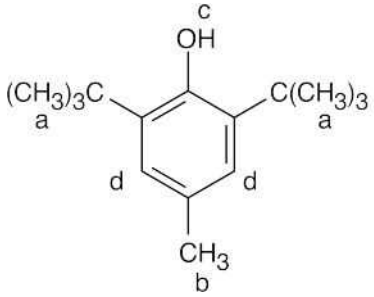
(f), (g)



a = 0.9 δ
b = 1.0 δ
c = 1.2 δ
d = 1.4 δ

2-Methyl-2-butanol

17.56

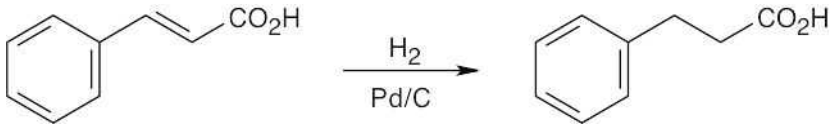


a = 1.41 δ
b = 2.24 δ
c = 5.00 δ
d = 6.97 δ

General Problems

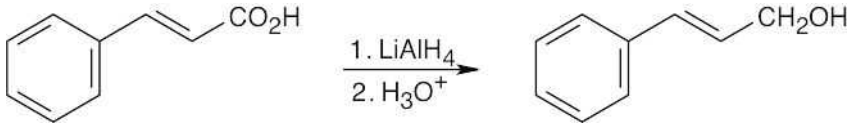
17.57 In these compounds you want to reduce some, but not all, of the functional groups present. To do this, choose the correct reducing agent.

(a)



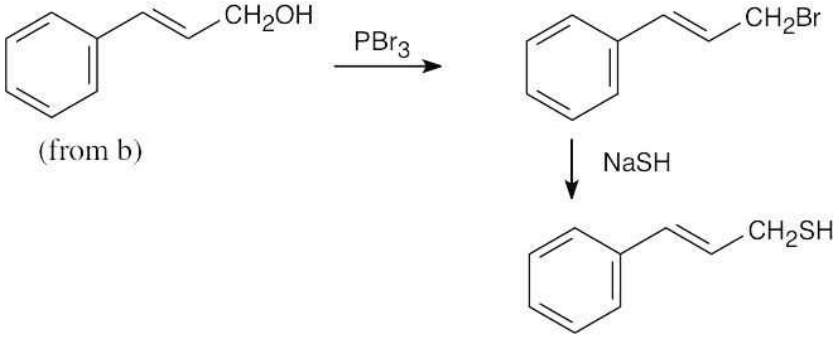
H_2 with a palladium catalyst hydrogenates carbon–carbon double bonds without affecting carbonyl double bonds.

(b)

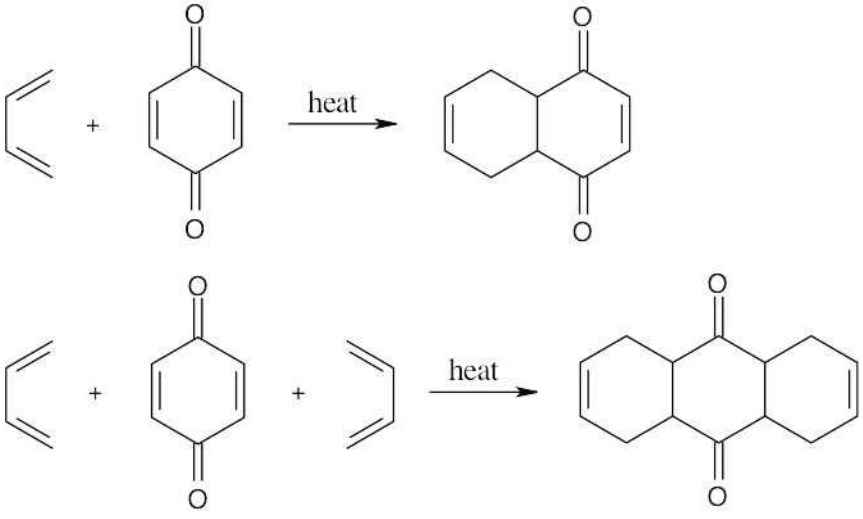


$LiAlH_4$ reduces carbonyl groups without affecting carbon–carbon double bonds.

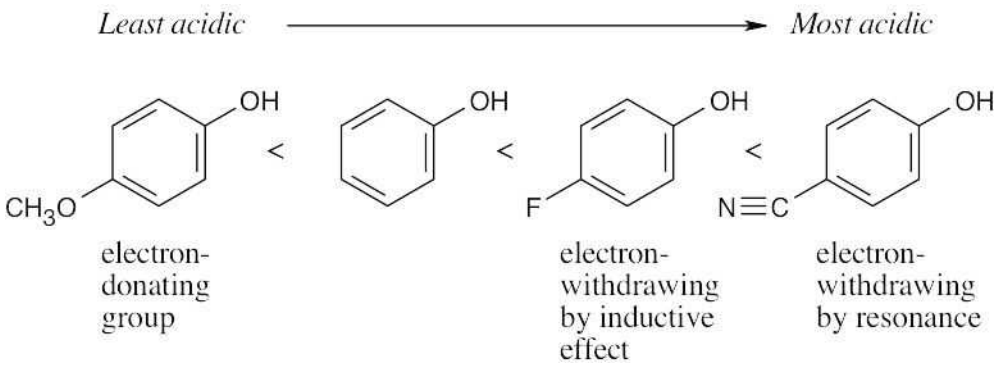
(c)



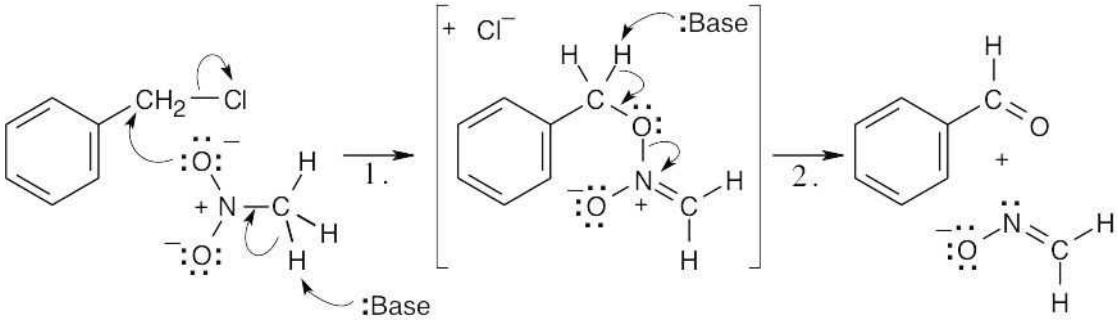
17.58



17.59 Remember that electron-withdrawing groups stabilize phenoxide anions and increase acidity. Electron-donating groups decrease phenol acidity.



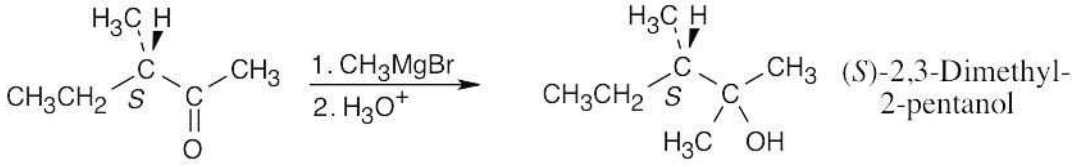
17.60



Step 1: S_N2 substitution.

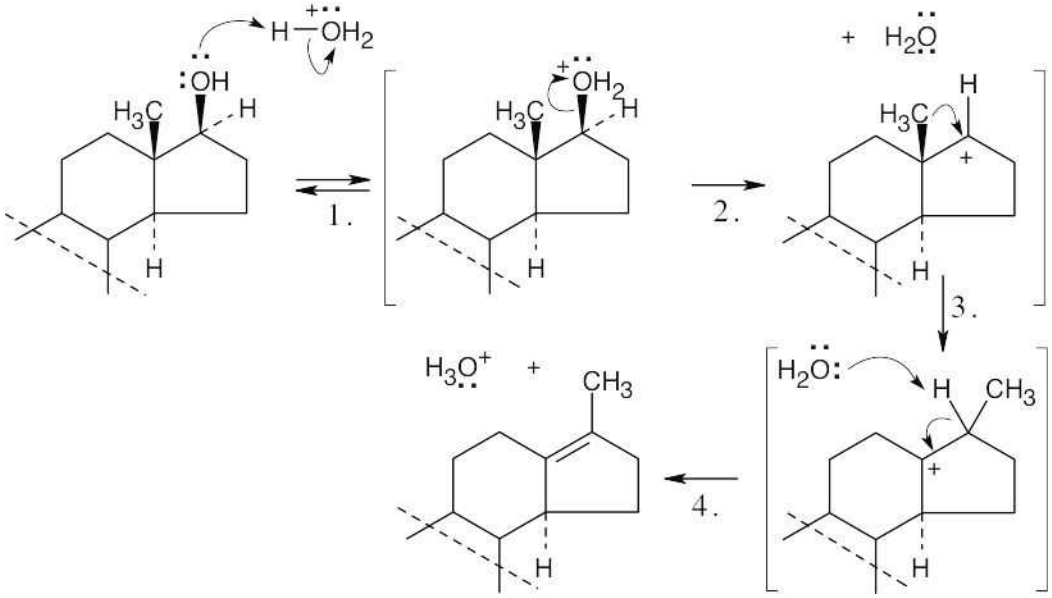
Step 2: E2 elimination.

17.61



Despite this problem's resemblance to Problem 17.24, the stereochemical outcome is different. Addition of methylmagnesium bromide to the carbonyl group doesn't produce a new chirality center and doesn't affect the chirality center already present. The product is pure (*S*)-2,3-dimethyl-2-pentanol, which is optically active.

17.62



Step 1: Protonation.

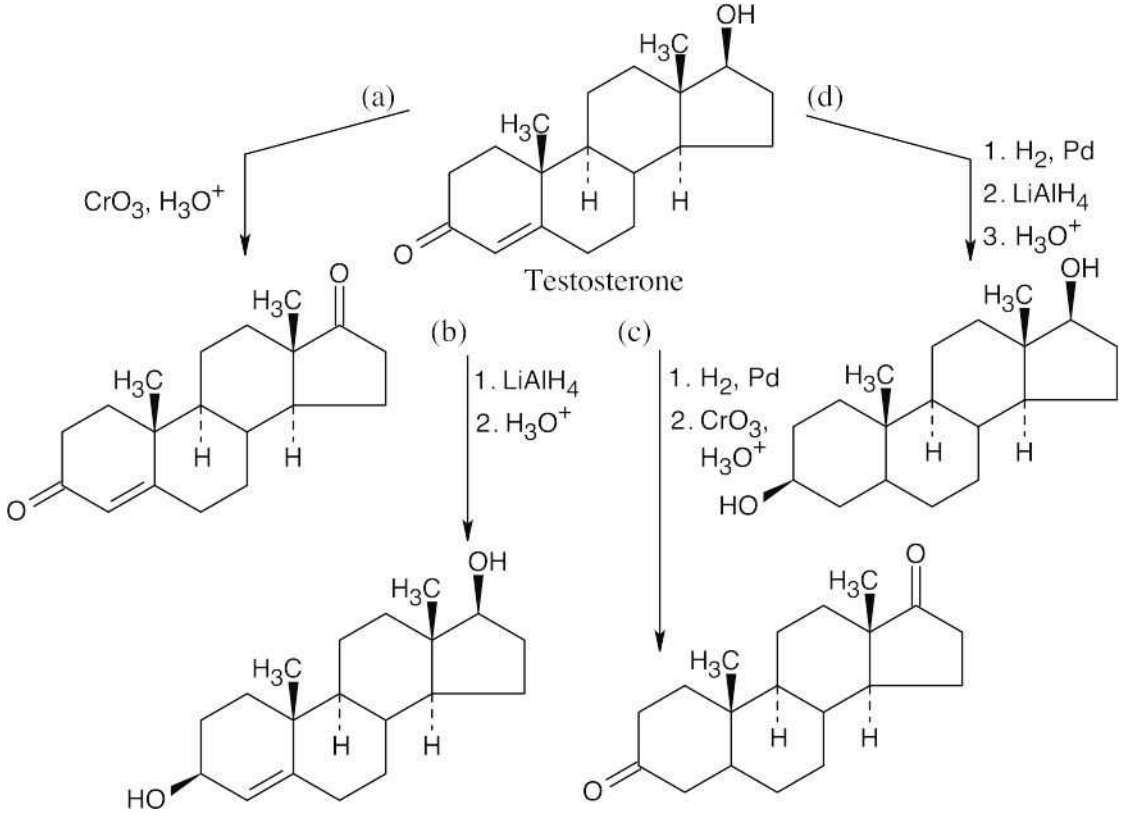
Step 2: Loss of H_2O.

Step 3: Alkyl shift.

Step 4: Loss of H^+.

This is a carbocation rearrangement involving the shift of an alkyl group. The sequence of steps is the same as those seen in Problems 17.27 and 17.28.

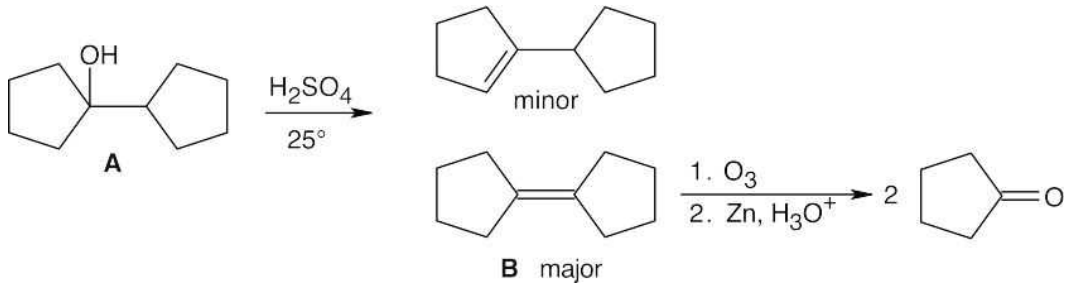
17.63



All of these transformations require the proper sequence of oxidations and reductions. In (d), $NaBH_4$ can also be used for reduction.

17.64 A phenoxide anion is stabilized by the electron-withdrawing resonance effect of a *p*-nitro group. Methyl groups ortho to the phenol have no effect on acidity, but the methyl groups that flank the nitro group of the 3,5-isomer force the nitro group out of planarity with the ring and reduce orbital overlap with the π orbitals of the ring. The resonance stabilization of the nitro group is reduced, and the pK_a of the phenol becomes higher, indicating lower acidity.

17.65



17.66 (a) Compound **A** has one double bond or ring.

(b) The infrared absorption at 3400 cm^{-1} indicates the presence of an alcohol. The weak absorption at 1640 cm^{-1} is due to a C=C stretch.

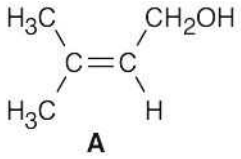
(c) (1) The absorptions at 1.63 δ and 1.70 δ are due to unsplit methyl protons. Because the absorptions are shifted slightly downfield, the protons are adjacent to an unsaturated center.

(2) The broad singlet at 3.83 δ is due to an alcohol proton.

(3) The doublet at 4.15 δ is due to two protons bonded to a carbon bearing an electronegative atom (oxygen, in this case).

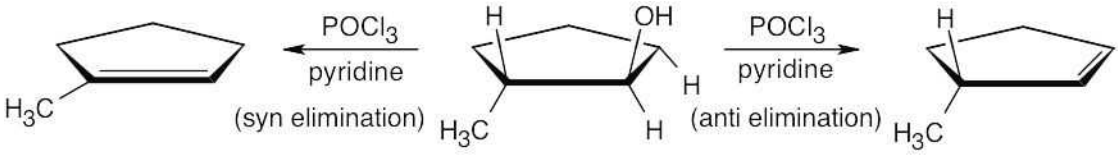
(4) The proton absorbing at 5.70 δ is a vinylic proton.

(d)



3-Methyl-2-buten-1-ol

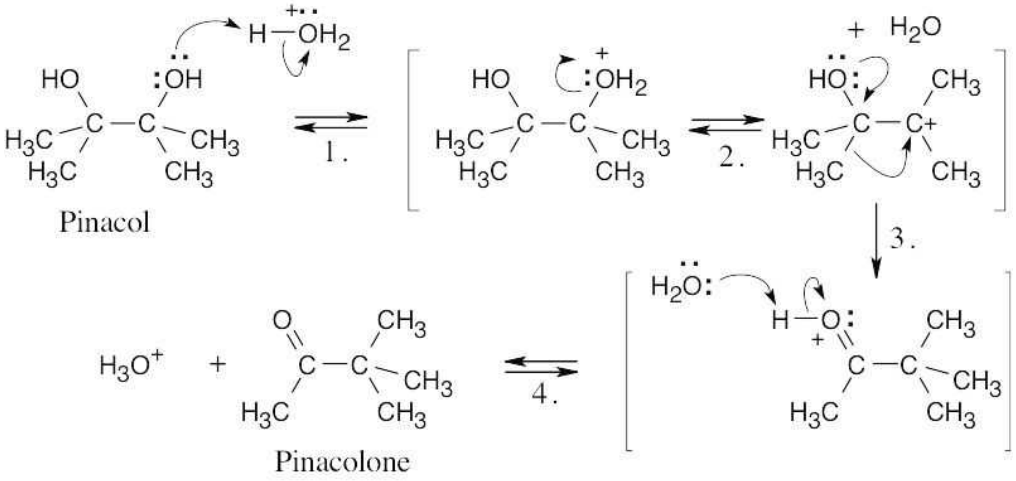
17.67



1-Methylcyclopentene *trans*-2-Methylcyclopentanol 3-Methylcyclopentene

The more stable dehydration product is 1-methylcyclopentene, which can be formed only via syn elimination. The major product of anti elimination is 3-methylcyclopentene. Since this product predominates, the requirement of anti periplanar geometry must be more important than formation of the more stable product.

17.68 The pinacol rearrangement follows a sequence of steps similar to other rearrangements we have studied in this chapter. The second hydroxyl group assists in the alkyl shift.



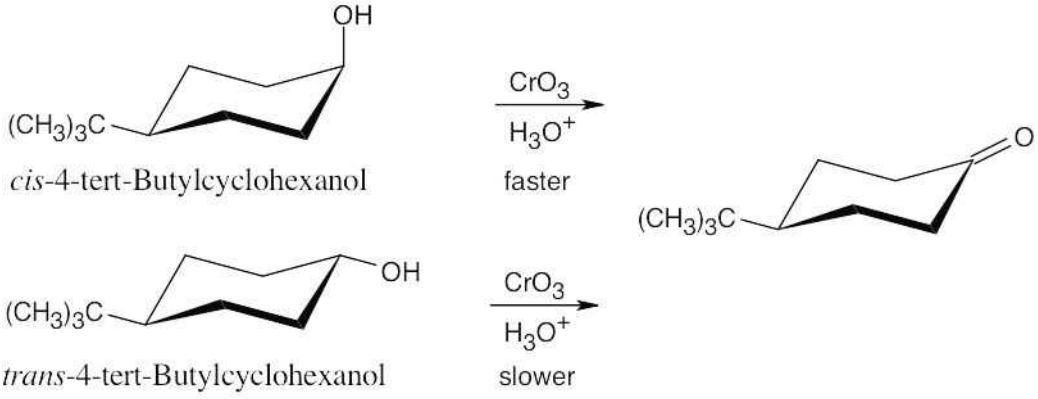
Step 1: Protonation.

Step 2: Loss of H_2O.

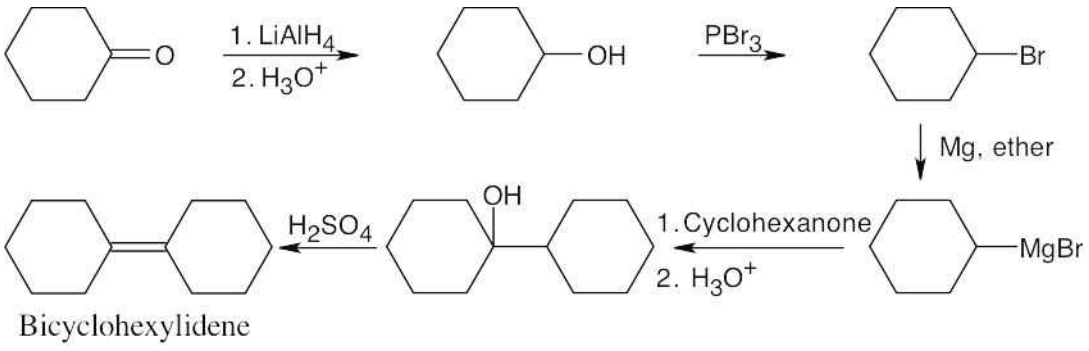
Step 3: Alkyl shift.

Step 4: Loss of H^+.

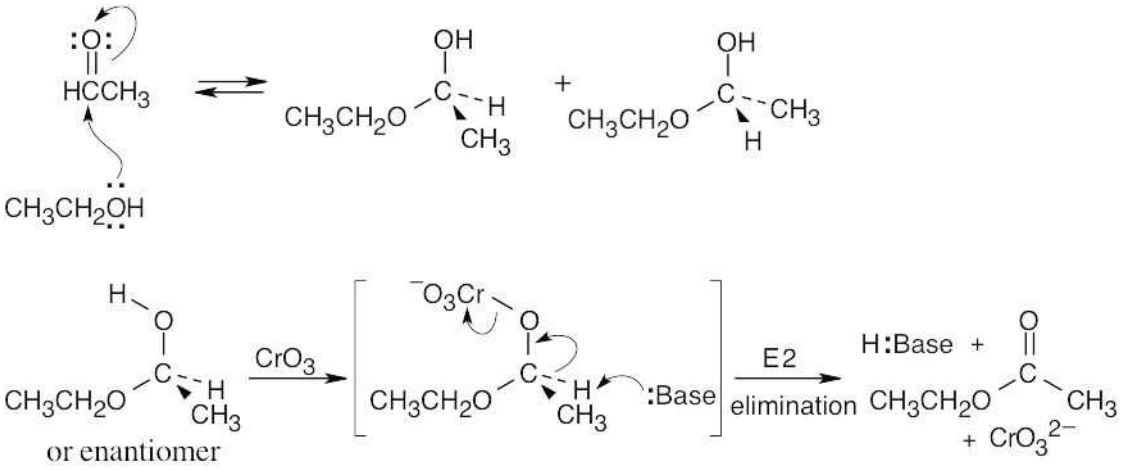
17.69 The hydroxyl group is axial in the cis isomer, which is expected to oxidize faster than the trans isomer. (Remember that the bulky *tert*-butyl group is always equatorial in the more stable isomer.)



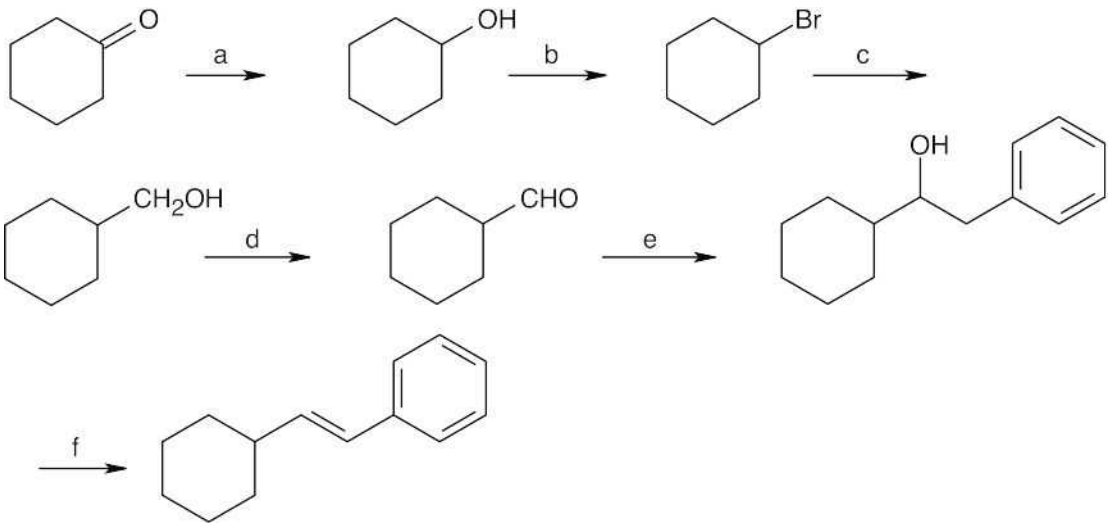
17.70



17.71 An alcohol adds to an aldehyde by a mechanism that we will study in a later chapter. The hydroxyl group of the addition intermediate undergoes oxidation (as shown in Section 17.7), and an ester is formed.



17.72

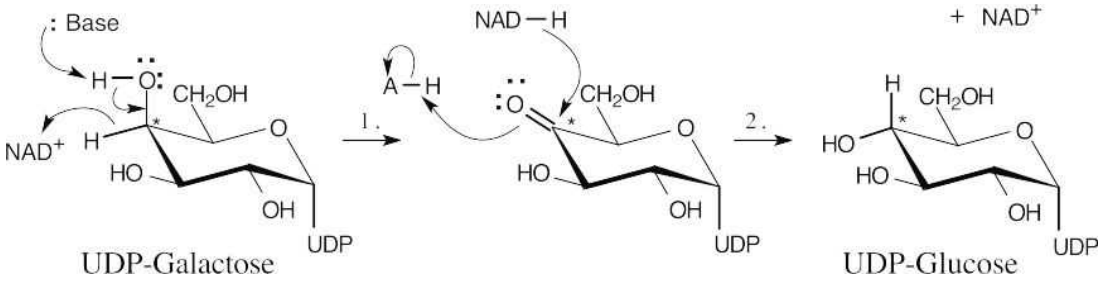


(a) $NaBH_4$, then H_3O^+ (b) PBr_3 (c) Mg, ether, then CH_2O

(d) Dess–Martin periodinane, CH_2Cl_2 (e) $C_6H_5CH_2MgBr$, then H_3O^+

(f) $POCl_3$, pyridine

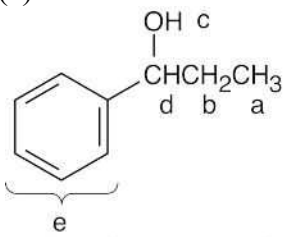
17.73



Step 1: Base deprotonates the C4 hydroxyl group while NAD^+ oxidizes the alcohol to a ketone.

Step 2: When the ketone is reduced by the NADH formed in Step 1, the configuration at the starred carbon is inverted, and UDP-glucose is formed.

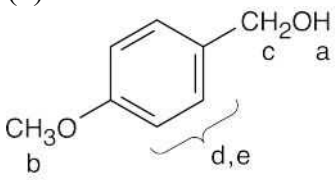
17.74 (a)



a = 0.88 δ
b = 1.8 δ
c = 2.32 δ
d = 4.54 δ
e = 7.24 δ

1-Phenyl-1-propanol

(b)



a = 2.60 δ
b = 3.76 δ
c = 4.53 δ
d = 6.85 δ
e = 7.23 δ

p-Methoxybenzyl alcohol

17.75 *Structural formula*: $C_8H_{10}O$ contains 4 multiple bonds and/or rings.

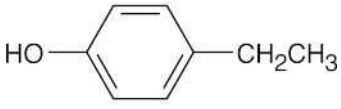
Infrared: The broad band at 3500 cm^{-1} indicates a hydroxyl group. The absorptions at 1500 cm^{-1} and 1600 cm^{-1} are due to an aromatic ring. The absorption at 830 cm^{-1} shows that the ring is *p*-disubstituted. Compound **A** is probably a phenol.

^1H NMR: The triplet at 1.16 δ (3 H) is coupled with the quartet at 2.55 δ (2 H). These two absorptions are due to an ethyl group.

The peaks at 6.74 δ-7.02 δ (4 H) are due to aromatic ring protons. The symmetrical splitting pattern of these peaks indicate that the aromatic ring is *p*-disubstituted.

The singlet absorption at 5.50 δ (1 H) is due to an –OH proton.

Compound **A**



p-Ethylphenol

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